



AKTU

B.Tech I-Year



Electronics Engg.

Unit-4 : Digital Electronics

in ONE SHOT

Pdf Notes + AKTU PYQs + Imp. Questions

By Gulshan Sir



FIRST YEAR

M-1

M-2

PHYSICS

ELECTRONICS



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SECOND YEAR

M-3

M-4

DSTL



By Gulshan

UNIT-4: Digital Electronics

- Number system & representation
- Binary arithmetic
- Introduction of Basic and Universal Gates
- Using Boolean algebra simplification of Boolean function
- K - Map Minimization up to 6 Variables

Number system : The most commonly used number systems in digital technology are

- (1) Binary
- (2) Decimal
- (3) Octal
- (4) Hexadecimal

Number System	Base / Radix	Digits Used	Example
Binary	2	0, 1	$(101111)_2$
Octal	8	0, 1, 2, 3, 4, 5, 6, 7	$(47525)_8$
Decimal	10	0, 1, 2, 3, 4, 5, 6, 7, 8, 9	$(316875)_{10}$
Hexadecimal	16	0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C, D, E, F	$(9B21A)_{16}$

NOTE : A = 10, B = 11, C = 12, D = 13, E = 14, F = 15

(1) Decimal to Binary

Example : $(319.6875)_{10} = (?)_2$

2	319	R
2	159	1
2	79	1
2	39	1
2	19	1
2	9	1
2	4	1
2	2	0
	1	0
		1

$$\begin{aligned}
 0.6875 \times 2 &= 1.3750 & 1 \\
 0.3750 \times 2 &= 0.7500 & 0 \\
 0.7500 \times 2 &= 1.50 & 1 \\
 0.50 \times 2 &= 1.00 & 1
 \end{aligned}$$

$$(319.6875)_{10} = (10011111.1011)_2$$

(2) Decimal to Octal : Example $(319.6875)_{10} = (?)_8$

8	319
8	39
	4

R

7

7

4



$$0.6875 \times 8 = 5.5 \quad 5$$

$$0.5 \times 8 = 4.0 \quad 4$$



$$(319.6875)_{10} = (477.54)_8$$

(3) Decimal to Hexadecimal : Example $(319.6875)_{10} = (?)_{16}$

16	319
16	19
	1

R

15 (F)

3

1



$$0.6875 \times 16 = 11.0000$$

11 (B)

$$(319.6875)_{10} = (13F.B)_{16}$$

(4) Other conversion : Example $(319.6875)_{10} = (?)_4$

4	319
4	79
4	19
4	4
	1

R

3

3

3

0

1

$$0.6875 \times 4 = 2.75$$

2

$$0.75 \times 4 = 3.0$$

3

$$(319.6875)_{10} = (10333.23)_4$$

(1) Binary to decimal : Example $(1011.11)_2 = (?)_{10}$

$$\begin{aligned}(1011.11)_2 &= 1 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 1 \times 2^0 + 1 \times 2^{-1} + 1 \times 2^{-2} \\&= 8 + 0 + 2 + 1 + 0.5 + 0.25 \\&= (11.75)_{10}\end{aligned}$$

(2) Octal to decimal : Example $(475.25)_8 = (?)_{10}$

$$\begin{aligned}(475.25)_8 &= 4 \times 8^2 + 7 \times 8^1 + 5 \times 8^0 + 2 \times 8^{-1} + 5 \times 8^{-2} \\&= 256 + 56 + 5 + 0.25 + 0.078125 \\&= (317.328125)_{10}\end{aligned}$$

(3) Hexadecimal to decimal : Example $(9B2.1A)_{16} = (?)_{10}$

$$(9B2.1A)_{16} = 9 \times 16^2 + B \times 16^1 + 2 \times 16^0 + 1 \times 16^{-1} + A \times 16^{-2}$$

$$= 9 \times 256 + 11 \times 16 + 2 + 0.0625 + 10 \times 0.00390625$$

$$= (2482.1015625)_{10}$$

(4) Other conversion : Example $(3102.12)_4 = (?)_{10}$

$$(3102.12)_4 = 3 \times 4^3 + 1 \times 4^2 + 0 \times 4^1 + 2 \times 4^0 + 1 \times 4^{-1} + 2 \times 4^{-2}$$

$$= 192 + 16 + 0 + 2 + 0.25 + 0.125$$

$$= (210.375)_{10}$$

(5) Other conversion : Example $(432)_5 = (?)_{10}$

$$(432)_5 = 4 \times 5^2 + 3 \times 5^1 + 2 \times 5^0$$

$$= 100 + 15 + 2$$

$$= (117)_{10}$$

(6) Example : Determine the values of base if (i) $(193)_x = (623)_8$ (ii) $24 + 17 = 40$

$$(193)_x = (623)_8$$

$$1 \times x^2 + 9 \times x^1 + 3 \times x^0 = 6 \times 8^2 + 2 \times 8^1 + 3 \times 8^0$$

$$x^2 + 9x + 3 = 403$$

$$x^2 + 9x - 400 = 0$$

$$x = 16$$

$$(24)_x + (17)_x = (40)_x$$

$$2 \times x^1 + 4 \times x^0 + 1 \times x^1 + 7 \times x^0 = 4 \times x^1 + 0 \times x^0$$

$$2x + 4 + x + 7 = 4x$$

$$x = 11$$





➤ Easy Method for some Number system conversion



Base

4

8

16

Group of binary digits

2

3

4

8 4 2 1

0 = 0 0 0 0

1 = 0 0 0 1

2 = 0 0 1 0

3 = 0 0 1 1

4 = 0 1 0 0

5 = 0 1 0 1

6 = 0 1 1 0

7 = 0 1 1 1

8 = 1 0 0 0

9 = 1 0 0 1

10 = 1 0 1 0

11 = 1 0 1 1

12 = 1 1 0 0

13 = 1 1 0 1

14 = 1 1 1 0

15 = 1 1 1 1

(1) Binary to octal conversion

Example : $(100\ 111\ 111.1011)_2 = (?)_8$

100	111	111	.	101	100
4	7	7		5	4

$$(100111111.1011)_2 = (477.54)_8$$

(2) Octal to binary conversion

Example : $(325.63)_8 = (?)_2$

3	2	5	.	6	3
011	010	101		110	011

$$(325.63)_8 = (011010101.110011)_2$$

$$= (11010101.110011)_2$$

(3) Binary to Hexadecimal conversion

Example : $(100\ 111111.1011)_2 = (?)_{16}$

0001	0011	1111	.	1011
1	3	F		B

$$(100111111.1011)_2 = (13F.B)_{16}$$

(4) Hexadecimal to binary conversion

Example : $(5A9.B4)_{16} = (?)_2$

5	A	9	.	B	4
0101	1010	1001		1011	0100

$$(5A9.B4)_{16} = (010110101001.10110100)_2$$

$$= (10110101001.101101)_2$$

(5) Octal to Hexadecimal conversion

Easy Method : Octal $\rightarrow$ Binary $\rightarrow$ Hexadecimal

Example : $(745)_8 = (?)_{16}$

STEP-1: OCTAL TO BINARY

7	4	5
111	100	101

$$(745)_8 = (111100101)_2$$

STEP-1: BINARY TO HEXADECIMAL

0001	1110	0101
1	E	5

$$(111100101)_2 = (1E5)_{16}$$

$$(745)_8 = (1E5)_{16}$$

(6) Hexadecimal to octal

Easy Method : Hexadecimal $\rightarrow$ Binary $\rightarrow$ Octal

Example : $(13F.B)_{16} = (?)_8$

STEP-1: HEXADECIMAL TO BINARY

1	3	F	.	B
0001	0011	1111	.	1011

$$(13F.B)_{16} = (10011111.1011)_2$$

STEP-2: BINARY TO OCTAL

100	111	111	.	101	100
4	7	7	.	5	4

$$(10011111.1011)_2 = (477.54)_8$$

$$(13F.B)_{16} = (477.54)_8$$

(7) Other conversion : Example $(1011)_2 = (?)_4$

$$\boxed{10} \quad \boxed{11}$$

$$2 \quad 3$$

$$(1011)_2 = (23)_4$$

(8) Other conversion : Example $(637)_9 = (?)_5$

$$(\quad)_9 \longrightarrow (\quad)_{10} \longrightarrow (\quad)_5$$

$$\begin{aligned} (637)_9 &= 6 \times 9^2 + 3 \times 9^1 + 7 \times 9^0 \\ &= 486 + 27 + 7 \\ &= (520)_{10} \end{aligned}$$

5	520	R 0 4 0 4
5	104	
5	20	
	4	

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$$(520)_{10} = (4040)_5$$

Hence

$$\boxed{(637)_9 = (4040)_5}$$

(1) Binary Addition

	SUM	CARRY
$0 + 0 = 0$	0	0
$0 + 1 = 1$	1	0
$1 + 0 = 1$	1	0
$1 + 1 = 0$	1	1
$1 + 1 + 1 = 1$	1	1

Note:

(i) $1 + 1 = 10$

SUM = 0

Carry = 1

(ii) $1 + 1 + 1 = 11$

SUM = 1

Carry = 1

Example - 1 : Add $(1110)_2$ and $(1100)_2$

$$\begin{array}{r}
 11 \\
 1110 \\
 + 1100 \\
 \hline
 11010
 \end{array}$$

Example -2 : Add $(11001)_2$ and $(10111)_2$

$$\begin{array}{r}
 1111 \\
 11001 \\
 + 10111 \\
 \hline
 110000
 \end{array}$$

Example - 3 : Add binary number $(1110.110)_2$ to $(1100.011)_2$

$$\begin{array}{r}
 1111 \\
 1110.110 \\
 + 1100.011 \\
 \hline
 11011.001
 \end{array}$$

Binary Arithmetic

(1) Binary Subtraction

Subtraction

Borrow

$$0 - 0 = 0$$

0

$$1 - 0 = 1$$

0

$$1 - 1 = 0$$

0

$$0 - 1 = 1$$

1

Note:

$$0 - 1 = 10 - 1$$

$$= 2 - 1$$

$$= 1$$

Example -1 : Subtract $(10110)_2$ from $(11011)_2$

$$\begin{array}{r} 0 \\ 1\cancel{1}011 \\ -10110 \\ \hline 00101 \\ \hline \end{array}$$

Example - 2 : Subtract $(00001)_2$ from $(11000)_2$

$$\begin{array}{r} 011 \\ 1\cancel{1}000 \\ -00001 \\ \hline 10111 \\ \hline \end{array}$$

Example -3 : Subtract $(0101)_2$ from $(1100)_2$

$$\begin{array}{r} 001 \\ \cancel{1}\cancel{1}00 \\ -0101 \\ \hline 0111 \\ \hline \end{array}$$

Example - 4 : Subtract $(010111)_2$ from $(100010)_2$

$$\begin{array}{r} 01110 \\ \cancel{1}\cancel{0}\cancel{0}\cancel{0}\cancel{1} \\ -010111 \\ \hline 001011 \\ \hline \end{array}$$

Complements are used in the digital computers in order to simplify the subtraction operation and for the logical manipulation

For Binary System

- (i) 1' S complement
- (ii) 2' S complement

1' S complement

Given Number

1 0 1 0

1 1 0 1 0 1 0 1

1 0 0 1

1' S complement

0 1 0 1

0 0 1 0 1 0 1 0

0 1 1 0

$$2' \text{ S complement} = 1' \text{ S complement} + 1$$

Given

1 0 1 1

1' S complement

0 1 0 0

2' S complement

0 1 0 0

+ 0 0 0 1

0 1 0 1

1 0 1 1 0 0 1 0

0 1 0 0 1 1 0 1

0 1 0 0 1 1 0 1

0 1 0 0 1 1 1 0

➤ Representation of positive and negative number using 1' S complement

Given Number

1' S complement

+ 5

0 1 0 1

↑
Sign

- 5

1 0 1 0

➤ Representation of positive and negative number using 2' S complement

Given Number

2' S complement

+ 5

0 1 0 1

- 5

1 0 1 0

1
—
1 0 1 1
—

Binary Subtraction 1'S complement

$$A - B = A + (-B)$$

STEP – 1 : Find the 1'S complement of the number to be subtracted

STEP – 2 : Perform addition

STEP – 3 : **CASE – 1 :** If the final carry is 1, then add it to the result obtained in step-2

CASE – 2 : If the final carry is zero, result obtain in step-2 is negative and in the 1'S complement form

Example – 1 : Subtract using 1'S complement $(1011)_2 - (0100)_2$

$$\text{Let } A = (1011)_2$$

$$B = (0100)_2$$

$$1'S \text{ complement of } B = (1011)_2$$

$$\begin{array}{r} 1011 \\ 1011 \\ \hline 1011 \\ 0110 \\ \hline 1 \leftarrow \text{Final carry} \\ 0111 \end{array}$$

$$(1011)_2 - (0100)_2 = (0111)_2$$

Example – 2 : Subtract using 1'S complement $(0100)_2 - (1001)_2$

Let $A = (0100)_2$ and $B = (1001)_2$

1'S complement of $B = (0110)$

Hence

$$(0100)_2 - (1001)_2 = -(0101)_2$$

$$= -(5)_{10}$$

Final carry →

$$\begin{array}{r}
 \boxed{0} \ 1 \ 0 \ 0 \\
 0 \ 1 \ 0 \ 0 \\
 0 \ 1 \ 1 \ 0 \\
 \hline
 1 \ 0 \ 1 \ 0 \\
 \hline
 \end{array}$$

Since final carry is 0, Result is negative

1'S complement of $(1010)_2 = (0101)_2$

$$A - B = A + (-B)$$

STEP - 1 : Find the 2' S complement of number to be subtracted

STEP - 2 : Perform addition

STEP - 3 : **CASE - 1 :** If the final carry is generated, then the result is positive and is in its true form

CASE - 2 : If the final carry is not generated, then the result is negative and is in its 2' S complement form.

Example - 1 : Subtract using 2' S complement $(1001)_2 - (0101)_2$

Let $A = (1001)_2$ and $B = (0101)_2$

1's complement of $B = (1010)_2$

2's complement of $B = 1010$

$$\begin{array}{r} 1 \\ 1010 \\ \hline 1011 \end{array}$$

Final carry

$$\begin{array}{r} \boxed{1}011 \\ 1001 \\ \hline 1011 \\ \hline 0100 \end{array}$$

$$(1001)_2 - (0101)_2 = (0100)_2$$

Example - 2 : Subtract using 2's complement $(0100)_2 - (1001)_2$

Let $A = (0100)_2$ and $B = (1001)_2$

1's complement of $B = (0110)_2$

2's complement of $B = 0110$

↓ Final carry

$$\begin{array}{r}
 \boxed{0}100 \\
 0100 \\
 0111 \\
 \hline
 1011
 \end{array}$$

$$\begin{array}{r}
 1 \\
 0111 \\
 \hline
 \hline
 \end{array}$$

Since, final carry is not generated, Result is negative and is in its 2's complement form

1's complement of $1011 = 0100$

2's complement of $1011 = 0100$

$$\begin{array}{r}
 1 \\
 0101 \\
 \hline
 \hline
 \end{array}$$

Hence

$$(0100)_2 - (1001)_2 = -(0101)_2$$

$$= -(5)_{10}$$

- Boolean Algebra is used to analyze and simplify digital (logic) circuits.
- Each variable in Boolean algebra has values either 0 (Low) or 1 (High).
- The basic operations used in Boolean algebra are:
 - (a) Logical addition
 - (b) Logical multiplication
 - (c) Complementation
- **Laws of Boolean Algebra**
 - (1) Commutative Law
 - (2) Associative Law
 - (3) Distributive Law
 - (4) AND Laws
 - (5) OR Laws
 - (6) Inversion Law
 - (7) Absorption Law
 - (8) De Morgan's Law

(1) Commutative Law

The commutative law states that changing the sequence of the variables does not have any effect on the output of a logic circuit.

$$(a) A \cdot B = B \cdot A$$

$$(b) A + B = B + A$$

(2) Associative Law

This law states that the order in which the logic operations are performed is irrelevant as their effect is the same.

$$(a) (A \cdot B) \cdot C = A \cdot (B \cdot C)$$

$$(b) (A + B) + C = A + (B + C)$$

(3) Distributive Law

$$A \cdot (B + C) = A \cdot B + A \cdot C$$

(4) AND Laws

These laws use the AND operation. Therefore, they are called AND laws.

$$(a) A \cdot 0 = 0$$

$$(b) A \cdot 1 = A$$

$$(c) A \cdot A = A$$

$$(d) A \cdot \bar{A} = 0$$

(5) OR Laws

These laws use the OR operation. Therefore, they are called OR laws.

$$(a) A + 0 = A$$

$$(b) A + 1 = 1$$

$$(c) A + A = A$$

$$(d) A + \bar{A} = 1$$

(6) Inversion Law

The inversion law states that double inversion of a variable results in the original variable itself

$$\bar{\bar{A}} = A$$

(7) Absorption Law

This law enables a reduction in a complicated expression to a simpler one by absorbing like terms.

$$A + AB = A$$

(8) De Morgan's Law

The two important laws have been suggested by De Morgan's.

(i) This law states that the complement of a product is equal to the addition of the complements.

$$\overline{AB} = \bar{A} + \bar{B}$$

(ii) This law states that the complement of a sum is equal to the product of the complements.

$$\overline{A+B} = \bar{A} \cdot \bar{B}$$

NOTE

$$A + BC = (A+B)(A+C)$$

Example – 1 : Simplify the expression $A + \bar{A}B + AB$ using Boolean algebra.

$$\begin{aligned} & A + \bar{A}B + AB \\ &= A + B(\bar{A} + A) \\ &= A + B \quad (\because \bar{A} + A = 1) \end{aligned}$$

Example – 2 : Simplify the Boolean function using Boolean algebra theorems

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$$\begin{aligned} & A'B'C' + A'BC' + AB'C' + ABC' \\ &= A'C'(B' + B) + AC'(B' + B) \\ &= A'C' + AC' \quad (\because B' + B = 1) \\ &= C'(A' + A) \\ &= C' \quad (\because A' + A = 1) \end{aligned}$$

$$F = XY + XY'Z + YZ'$$

$$F = XY(Z + Z') + XY'Z + YZ'$$

$$F = XYZ + XYZ' + XY'Z + YZ'$$

$$F = XZ(Y + Y') + YZ'(X + 1)$$

$$F = XZ + YZ'$$

$$\left. \begin{array}{l} \because Y + Y' = 1 \\ X + 1 = 1 \end{array} \right\}$$

$$X + 1 = 1$$

Example – 4 : Simplify the following expression $Y = (A+B)(A+C)$.

$$Y = (A+B)(A+C)$$

$$Y = AA + AC + BA + BC$$

$$Y = A + AC + BA + BC \quad (\because AA = A)$$

$$Y = A(1+C) + BA + BC$$

$$Y = A + BA + BC \quad (\because 1+C = 1)$$

$$Y = A(1+B) + BC \quad (\because 1+B = 1)$$

$$Y = A + BC$$

Any logic expression may be expressed in the following two forms

(i) Sum of Product form (SOP)

$$f(A, B, C) = AB + BC + AC$$

Standard or Canonical SOP form

Each product term consists of all the variables in the complemented or un-complemented form.

$$f(A, B, C) = ABC + A\bar{B}\bar{C} + \bar{A}BC$$

(ii) Product of Sum form (POS)

$$f(A, B, C) = (A+B).(B+C).(A+C)$$

Standard or Canonical POS form

Each sum term consists of all the variables in the complemented or un-complemented form.

$$f(A, B, C) = (A + B + C) . (A + \bar{B} + \bar{C}) . (\bar{A} + B + C)$$

Conversion of a logical expression to standard SOP or POS form

Example – 1 : Convert the expression $Y = A B + \bar{A} C + B C$ into the standard SOP form.

$$Y = AB + A\bar{C} + BC$$

$$Y = AB(C + \bar{C}) + A\bar{C}(B + \bar{B}) + BC(A + \bar{A}) \quad \left\{ \because A + \bar{A} = 1 \right\}$$

$$Y = ABC + AB\bar{C} + A\bar{C}B + A\bar{C}\bar{B} + BCA + B\bar{C}\bar{A}$$

$$Y = ABC + ABC + AB\bar{C} + A\bar{C}B + A\bar{C}\bar{B} + \bar{A}BC$$

$$Y = ABC + AB\bar{C} + A\bar{C}B + \bar{A}BC \quad \left\{ \because A + A = A \right\}$$

Example – 2 : Convert the expression $Y = (A+B)(A+C)(B+\bar{C})$ to standard POS form.

$$Y = (A+B)(A+C)(B+\bar{C})$$

$$Y = (A+B+C\bar{C})(A+C+B\bar{B})(B+\bar{C}+A\bar{A})$$

$$Y = (A+B+C)(A+B+\bar{C})(A+C+B)(A+C+\bar{B})(B+\bar{C}+A)(B+\bar{C}+\bar{A})$$

$$Y = (A+B+C)(A+B+\bar{C})(A+\bar{B}+C)(\bar{A}+B+\bar{C})$$

Minterm

- A product term which contains all the function variables either in complemented form or in un-complemented form is called a Minterm.
- Each individual term in the standard SOP form is called a Minterm

Maxterm

- A sum term which contains all the function variables either in complemented form or in un-complemented form is called a Maxterm.
- Each individual term in the standard POS form is called a Maxterm

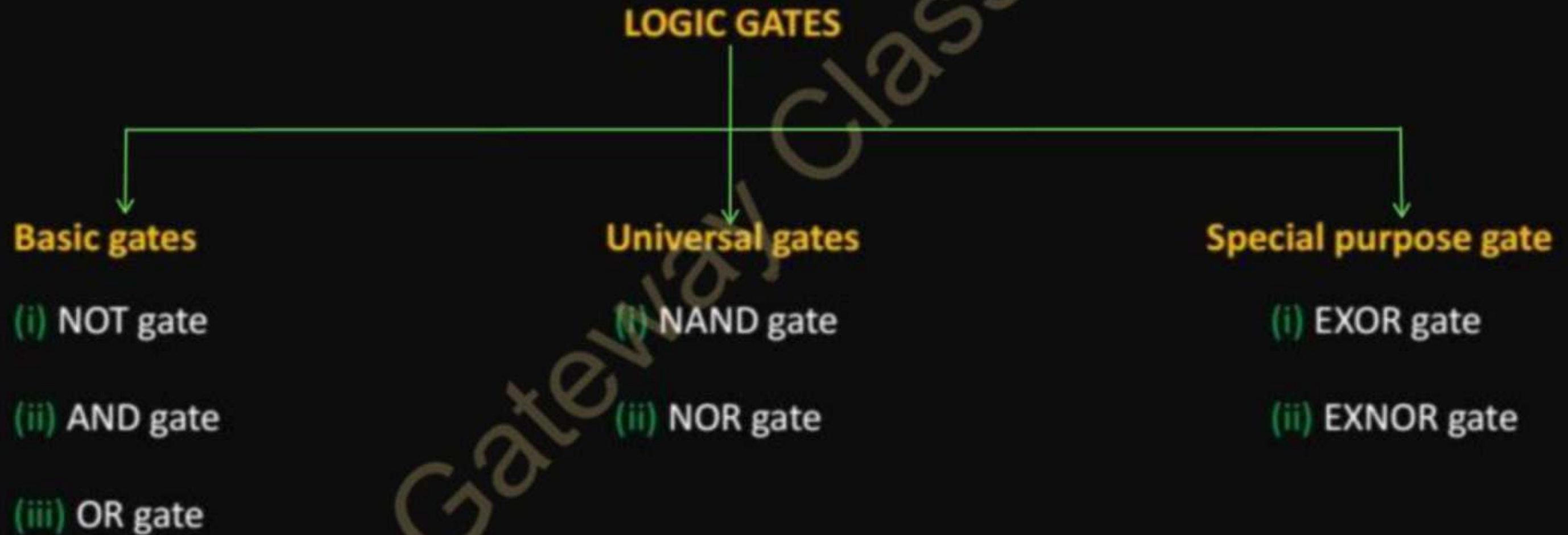
Maximum number of minterms = 2^n

Maximum number of maxterms = 2^n

Where n = number of variables

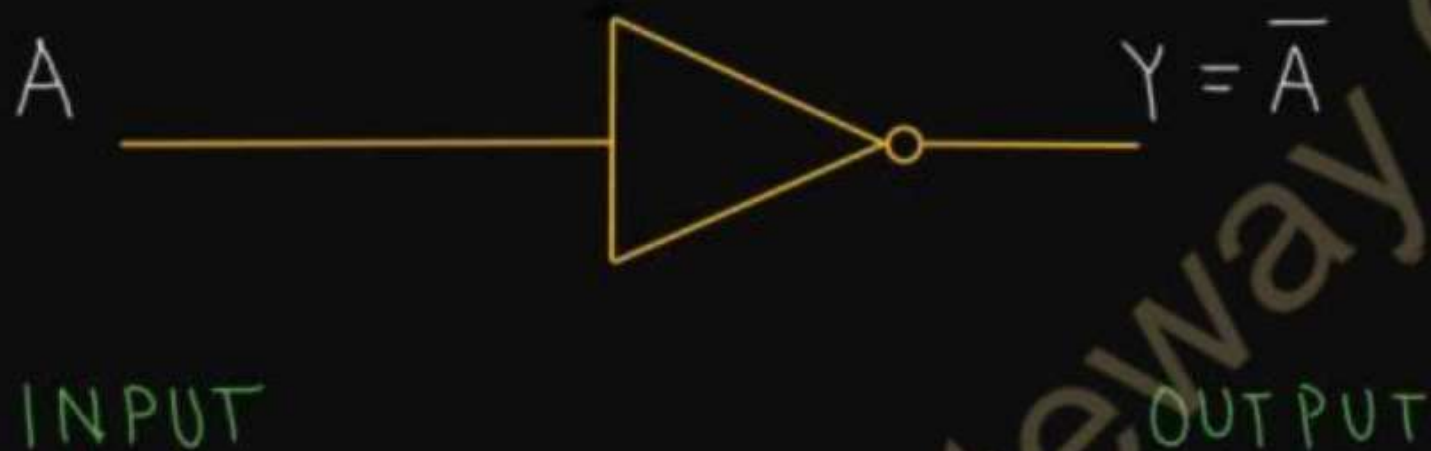
Variables		Minterms	Maxterms
A	B	m_i	M_i
0	0	$m_0 = \bar{A} \bar{B}$	$M_0 = A + B$
0	1	$m_1 = \bar{A} B$	$M_1 = A + \bar{B}$
1	0	$m_2 = A \bar{B}$	$M_2 = \bar{A} + B$
1	1	$m_3 = AB$	$M_3 = \bar{A} + \bar{B}$

- The most basic element of digital circuits are the logic gates.
- Boolean functions may be practically implemented by using Logic gates.
- Electronic gates require a power supply.



- The NOT gate is also known as an inverter.
- Output is the inverted version or complement of its input.
- NOT gate has only one input and one output.

Symbol



Truth Table

Input	Output
A	$Y = \bar{A}$
0	1
1	0

- The AND gate is an electronic circuit that gives a high output (1) only if all its inputs are high.
- It performs logical multiplication on its inputs.

Symbol



INPUT

OUTPUT

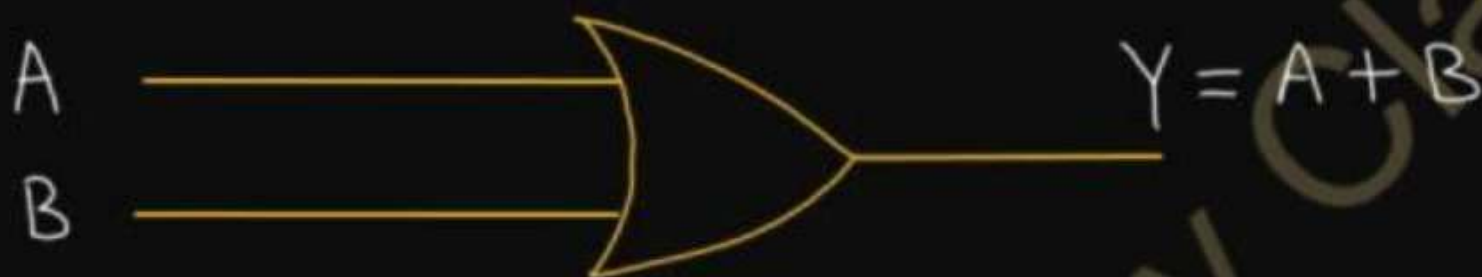
Truth Table

Input		Output
A	B	$Y = A . B$
0	0	0
0	1	0
1	0	0
1	1	1

(3) OR gate

- The OR gate is an electronic circuit that gives a high output (1) if one or more of its inputs are high.
- It performs logical addition on its inputs.

Symbol



INPUT

OUTPUT

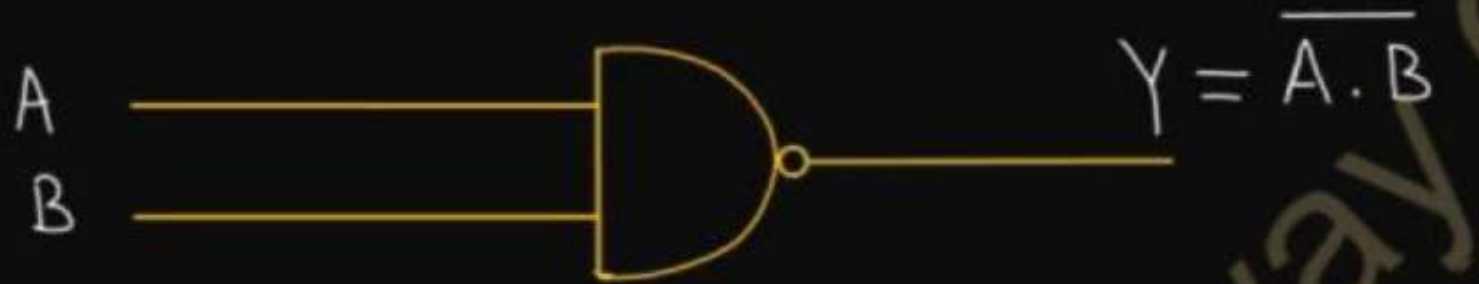
Truth Table

Input		Output
A	B	$Y = A + B$
0	0	0
0	1	1
1	0	1
1	1	1

(4) NAND gate

- NOT + AND gate
- The input of all NAND gates are high if any of the input are low
- The small circle represents inversion

Symbol



INPUT

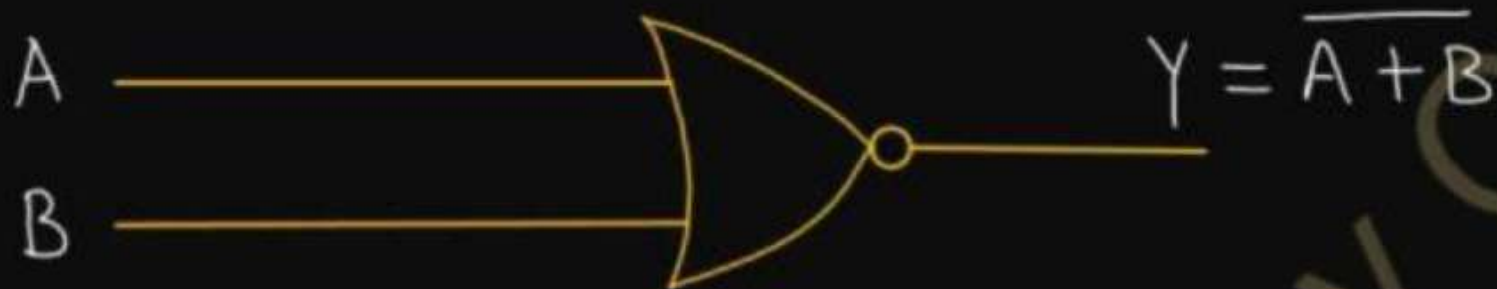
OUTPUT

Truth Table

Input		Output
A	B	$Y = \overline{A \cdot B}$
0	0	1
0	1	1
1	0	1
1	1	0

- Not + OR Gate
- The outputs of all NOR gates are low if any of the inputs are high.

Symbol



INPUT

OUTPUT

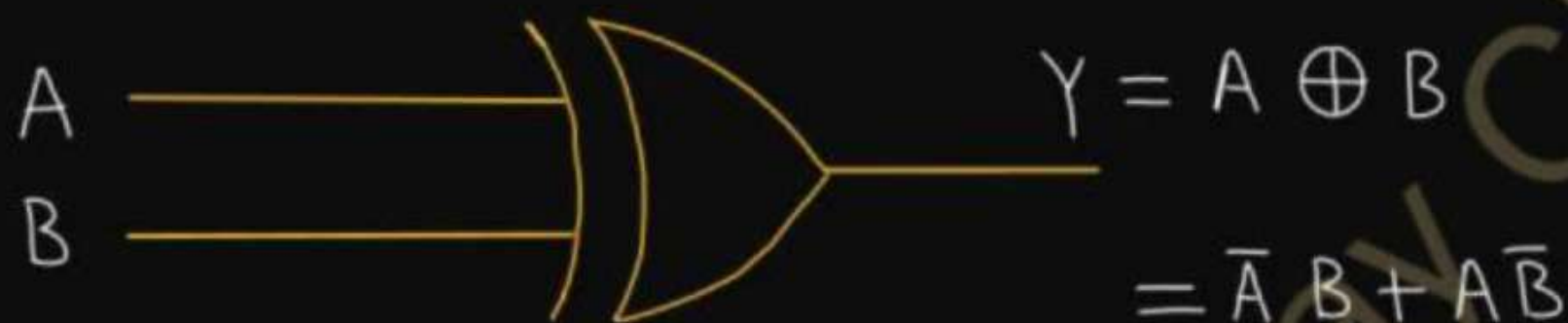
Truth Table

Input		Output
A	B	$Y = \overline{A + B}$
0	0	1
0	1	0
1	0	0
1	1	0

(6) EXOR gate

- The EXOR gate is a circuit which will give a high output if both the input are different
- $\oplus$ is used to show EXOR operation

Symbol



INPUT

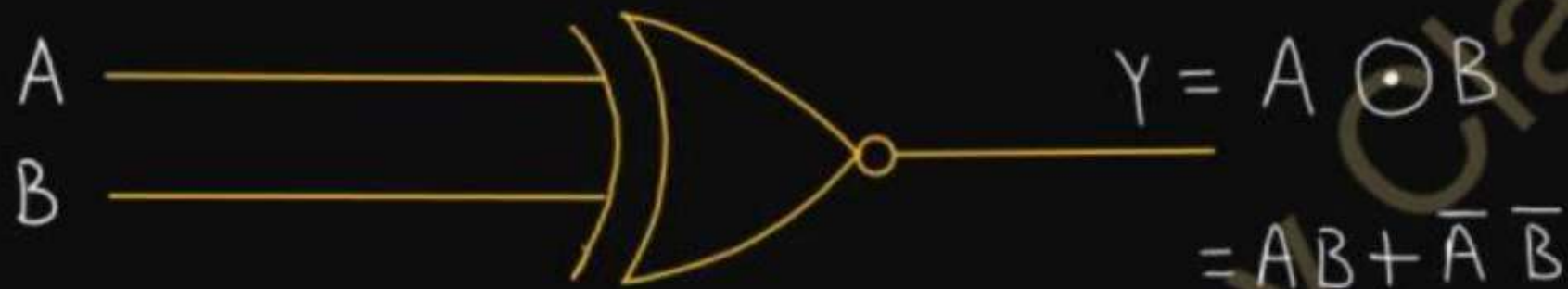
OUTPUT

Truth Table

Input		Output
A	B	$Y = A \oplus B$
0	0	0
0	1	1
1	0	1
1	1	0

- The EX-NOR gate is a circuit which will give a high output if both the input are same.
- $\odot$ is used to show EX-NOR gate.

Symbol



INPUT

OUTPUT

Truth Table

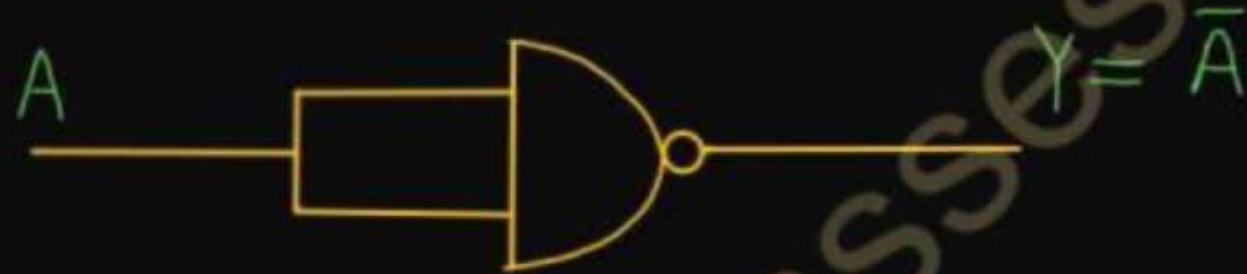
Input		Output
A	B	$Y = A \odot B$
0	0	1
0	1	0
1	0	0
1	1	1

- NAND and NOR gates are called as universal gates because it is possible to implement any Boolean expression with the help of only NAND gates or only NOR gates.
- A user can build any logical gates or combinational circuit with the help of only NAND gates or only NOR gates.
- NAND and NOR gates are economical and easier to fabricate and are the basic gates used in all IC digital logic families.

(1) NOT gate using NAND gate

$$Y = \bar{A}$$

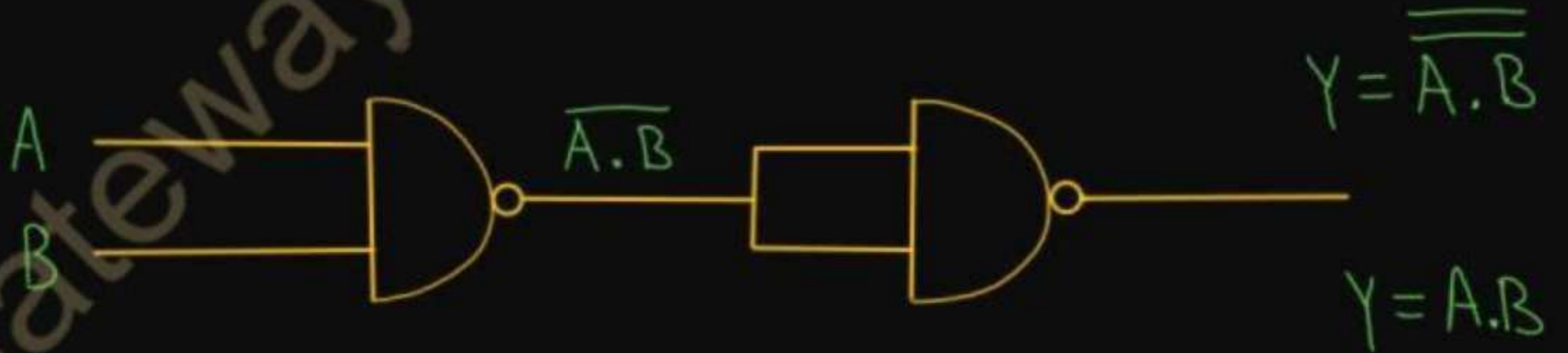
$$Y = \overline{A \cdot A}$$



(2) AND gate using NAND gate

$$Y = A \cdot B$$

$$Y = \overline{\overline{A \cdot B}}$$



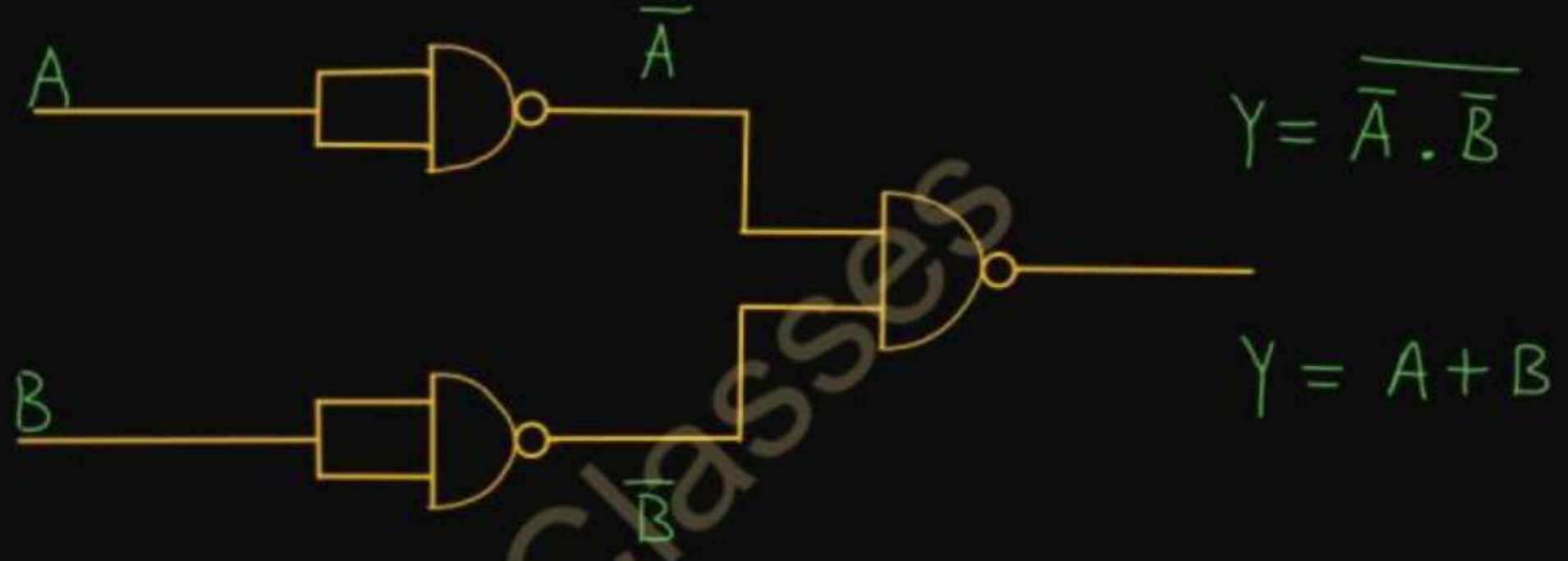
(3) OR gate using NAND gate

$$Y = A + B$$

$$Y = \overline{\overline{A + B}}$$

$$Y = \overline{\overline{A} \cdot \overline{B}}$$

$$Y = A + B$$

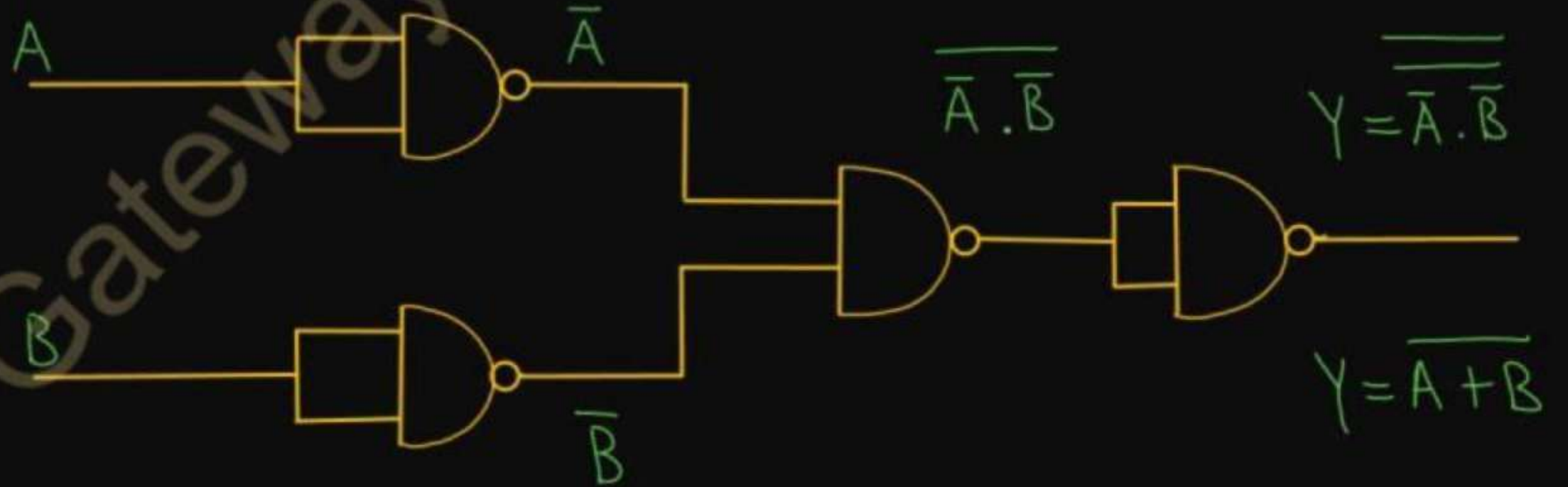


(4) NOR gate using NAND gate

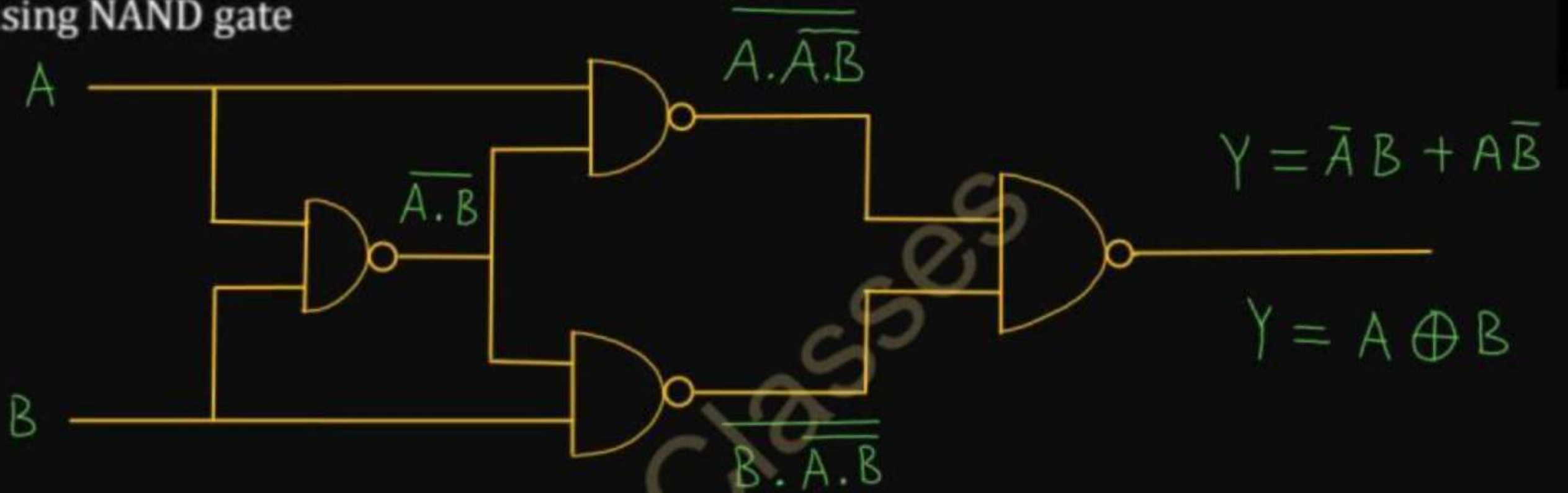
$$Y = \overline{A + B}$$

$$Y = \overline{\overline{\overline{A + B}}}$$

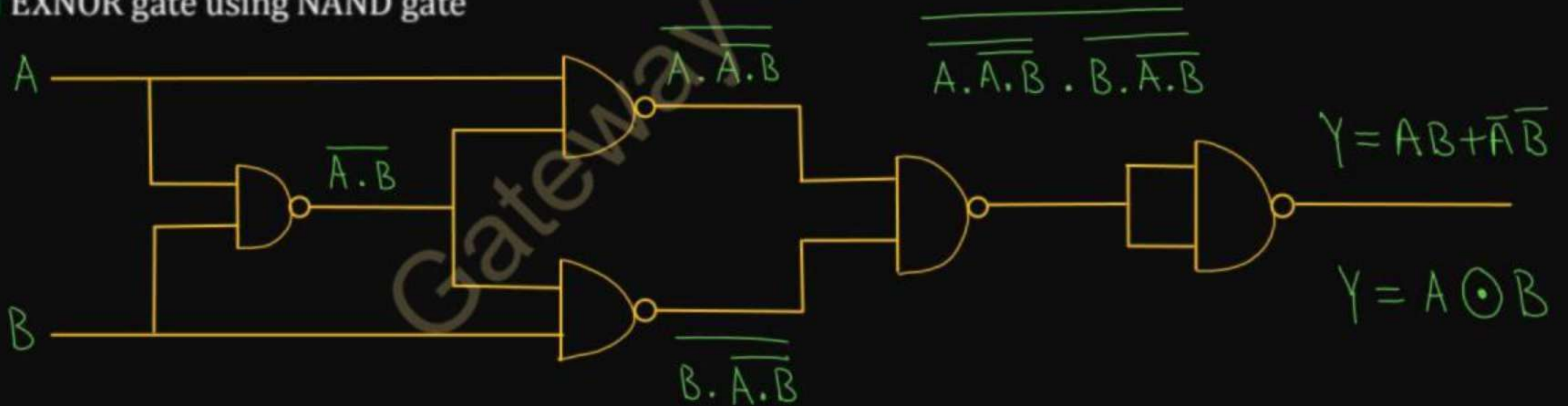
$$Y = \overline{\overline{A} \cdot \overline{B}}$$



(5) EXOR gate using NAND gate



(6) EXNOR gate using NAND gate



(1) NOT gate using NOR gate

$$Y = \bar{A}$$

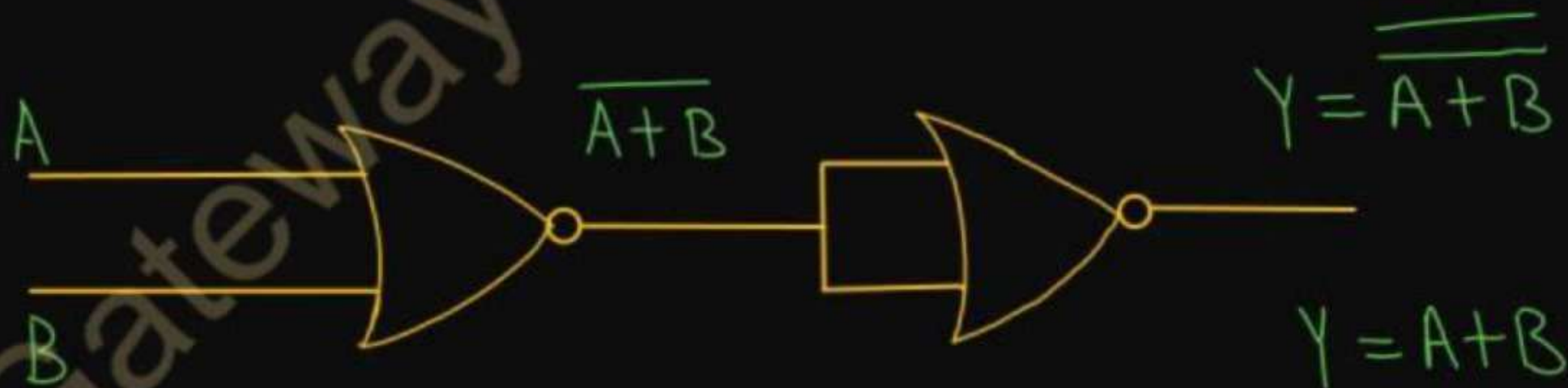
$$Y = \overline{A + A}$$



(2) OR gate using NOR gate

$$Y = A + B$$

$$Y = \overline{\overline{A + B}}$$

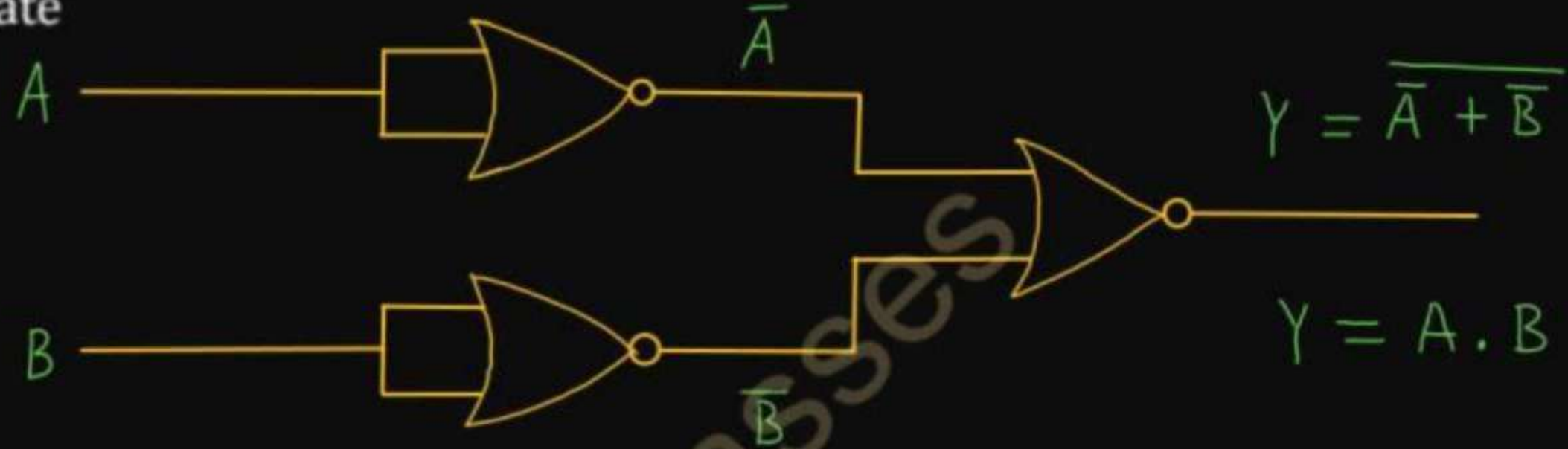


(3) AND gate using NOR gate

$$Y = A \cdot B$$

$$Y = \overline{\overline{A \cdot B}}$$

$$Y = \overline{\overline{A} + \overline{B}}$$

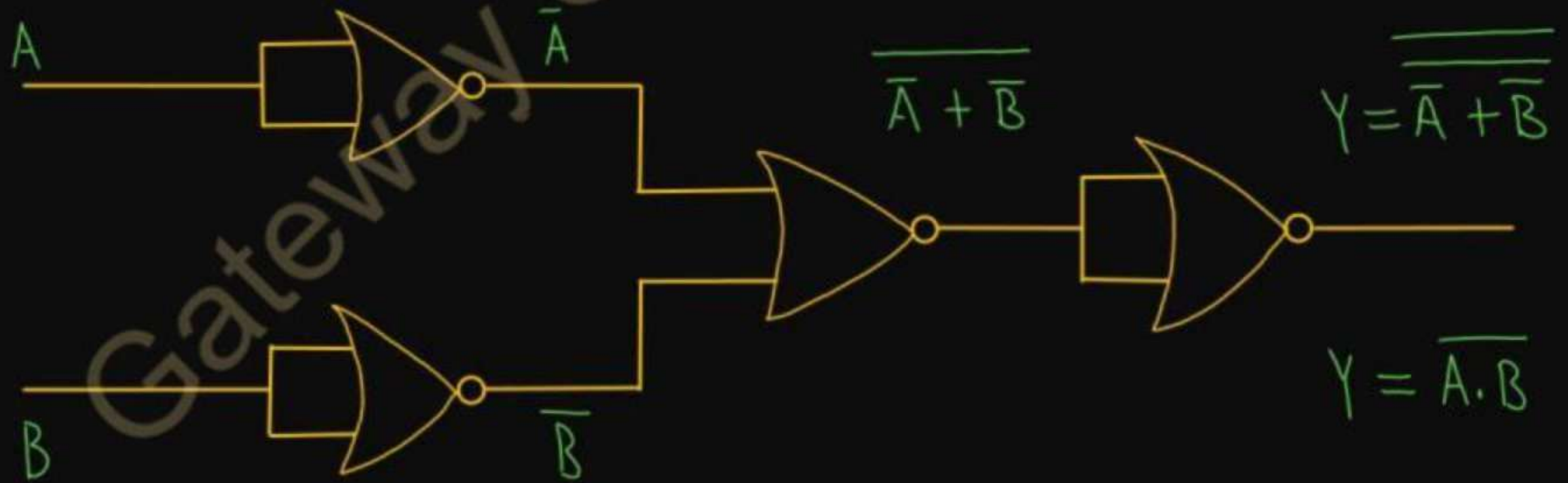


(4) NAND gate using NOR gate

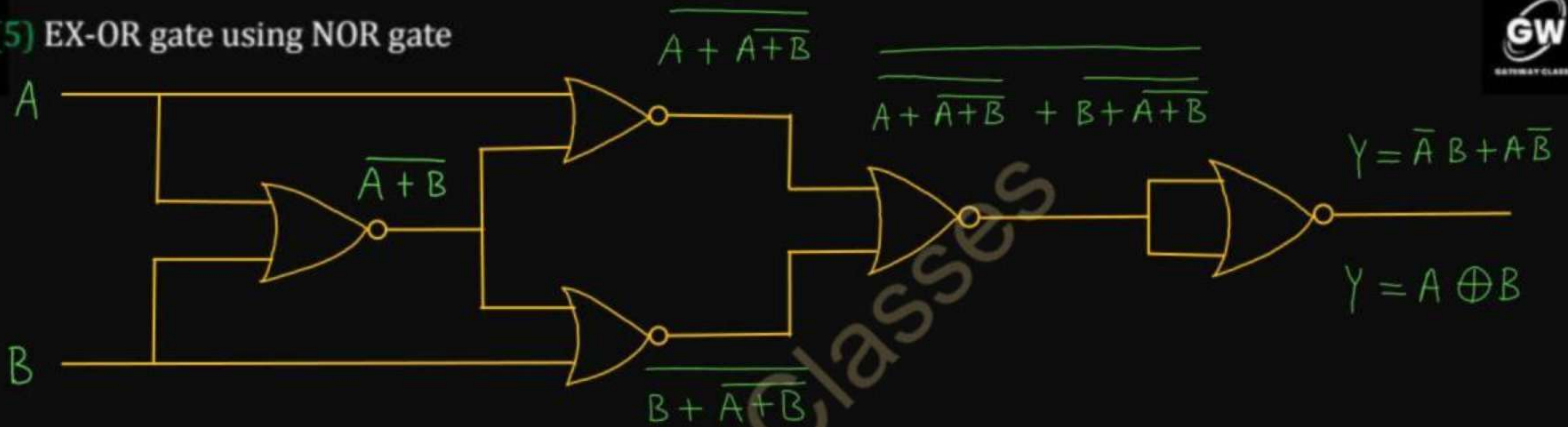
$$Y = \overline{A \cdot B}$$

$$Y = \overline{\overline{\overline{A \cdot B}}}$$

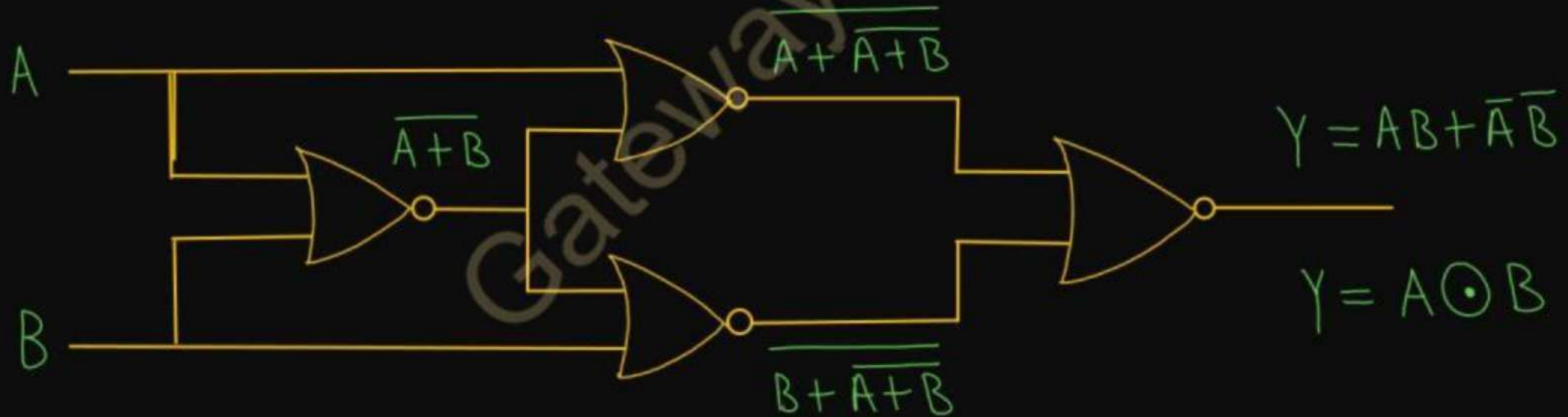
$$Y = \overline{\overline{\overline{A} + \overline{B}}}$$



(5) EX-OR gate using NOR gate



(6) EXNOR gate using NOR gate



- The K-map is a pictorial representation of truth table of the Boolean function and used to minimize the Boolean function
- K-map matrix contain 2^n cell, where n is the total number of variables

K-map

For SOP form

Each Cell contain a Minterm

- | | |
|---|-----------------------------|
| 0 | Variable with complement |
| 1 | Variable without complement |

For POS form

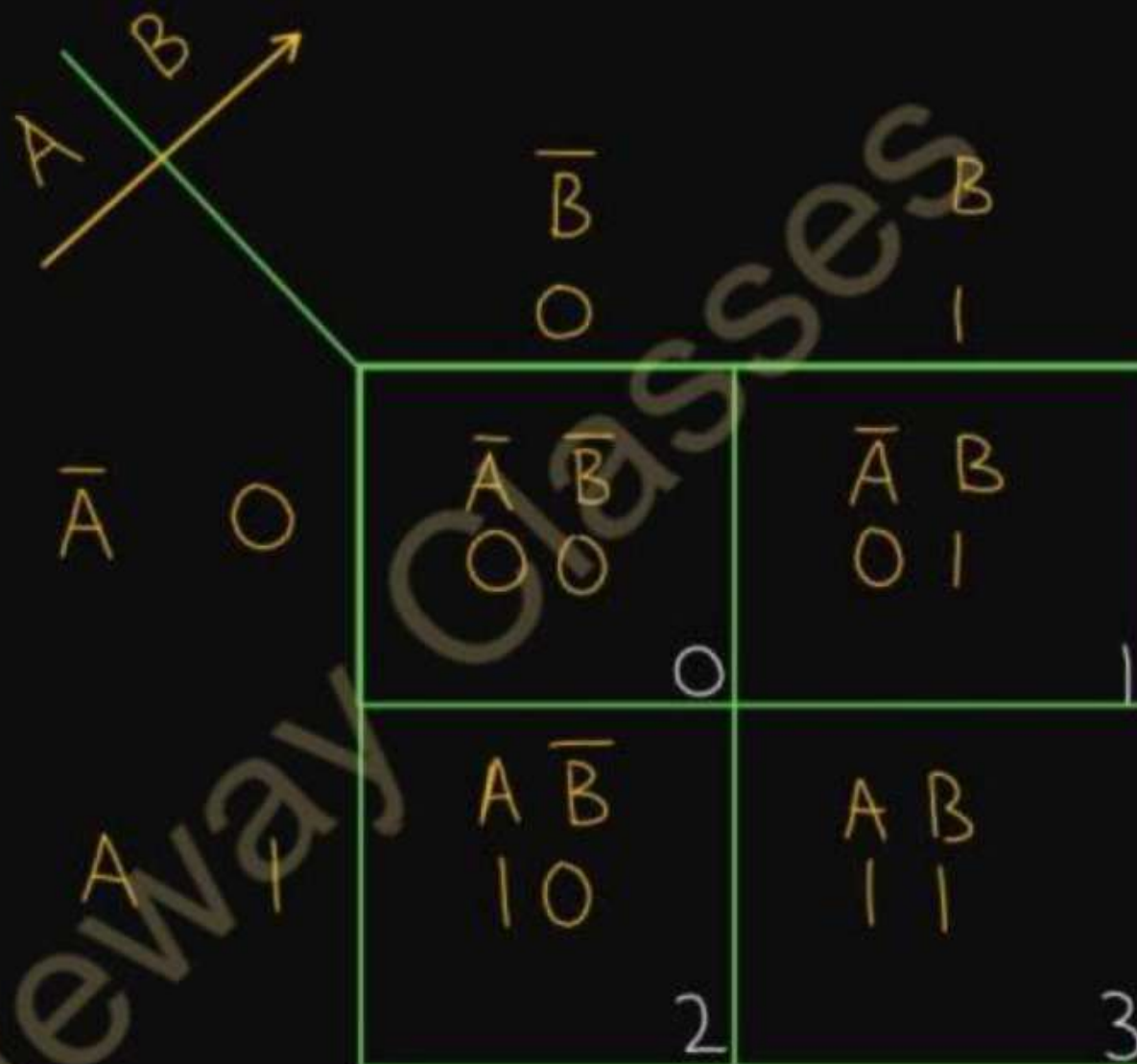
Each cell contain a Maxterm

- | | |
|---|-----------------------------|
| 0 | Variable without complement |
| 1 | Variable with complement |

K-Map for SOP Form

(1) For two variable A and B

- No of cells in K-map = $2^2 = 4$
- Each cell contain a Minterm



(2) For three variables A, B, C

- No of cells in K-map = $2^3 = 8$
- Each cell contain a Minterm

		BC			
		$\bar{B}\bar{C}$ 00	$\bar{B}C$ 01	BC 11	$B\bar{C}$ 10
A	$\bar{A}$ 0	$\bar{A}\bar{B}\bar{C}$ 000	$\bar{A}\bar{B}C$ 001	$\bar{A}B\bar{C}$ 011	$\bar{A}BC$ 010
	A 1	$A\bar{B}\bar{C}$ 100	$A\bar{B}C$ 101	$AB\bar{C}$ 111	ABC 110
		0	1	3	2
		4	5	7	6

(3) For four variables A, B, C, D

- No of cells in K-map = $2^4 = 16$
- Each cell contain a Minterm

		CD			
		$\bar{C}\bar{D}$ 00	$\bar{C}D$ 01	CD 11	$C\bar{D}$ 10
AB	$\bar{A}\bar{B}$ 00	$\bar{A}\bar{B}\bar{C}\bar{D}$ 0000 0	$\bar{A}\bar{B}\bar{C}D$ 0001 1	$\bar{A}\bar{B}CD$ 0011 3	$\bar{A}\bar{B}C\bar{D}$ 0010 2
	$\bar{A}B$ 01	$\bar{A}B\bar{C}\bar{D}$ 0100 4	$\bar{A}B\bar{C}D$ 0101 5	$\bar{A}BCD$ 0111 7	$\bar{A}BC\bar{D}$ 0110 6
	AB 11	$AB\bar{C}\bar{D}$ 1100 12	$AB\bar{C}D$ 1101 13	$ABCD$ 1111 15	$ABC\bar{D}$ 1110 14
	$A\bar{B}$ 10	$A\bar{B}\bar{C}\bar{D}$ 1000 8	$A\bar{B}\bar{C}D$ 1001 9	$A\bar{B}CD$ 1011 11	$A\bar{B}C\bar{D}$ 1010 10

Rule of Minimization

- Put 1 for each Minterm and 0 for each maxterm
- **Formation of Groups** : Form minimum number of groups and Form big size group

To Reduce terms

To Reduce variables

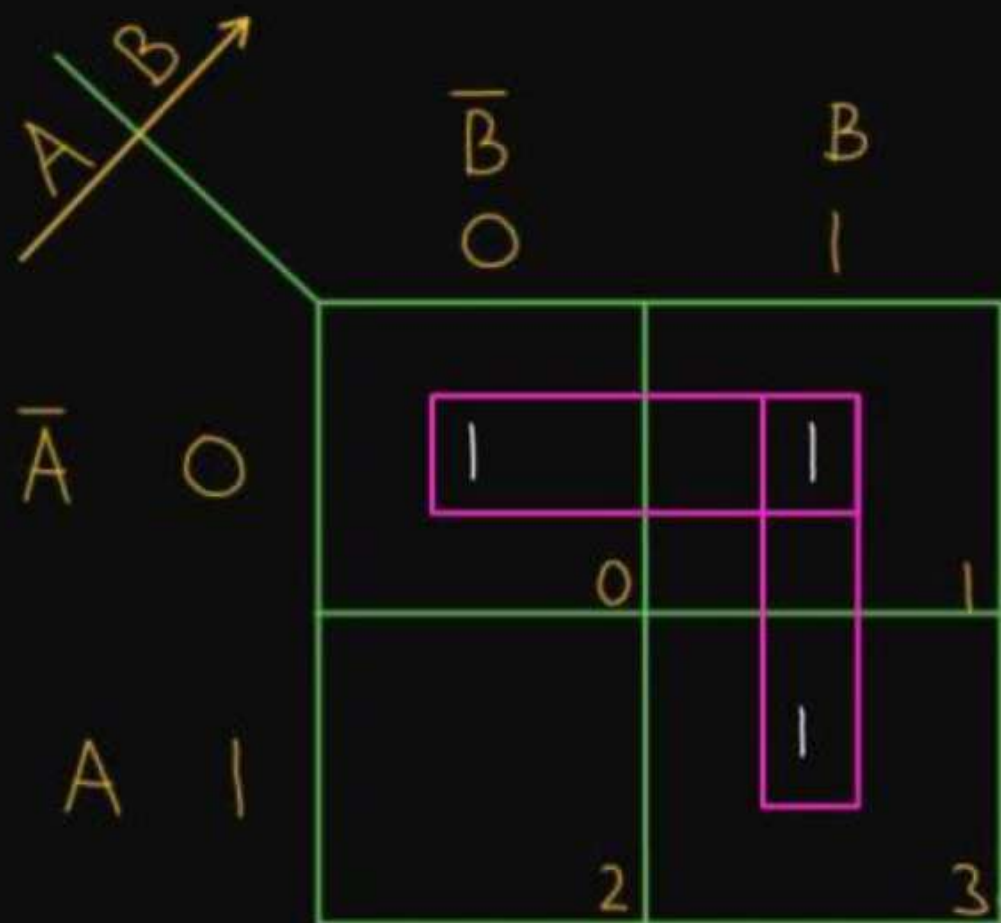
(1) 16 Group = 2^4 (4 Variables minimise)

(2) 8 Group = 2^3 (3 Variables minimise)

(3) 4 Group = 2^2 (2 Variables minimise)

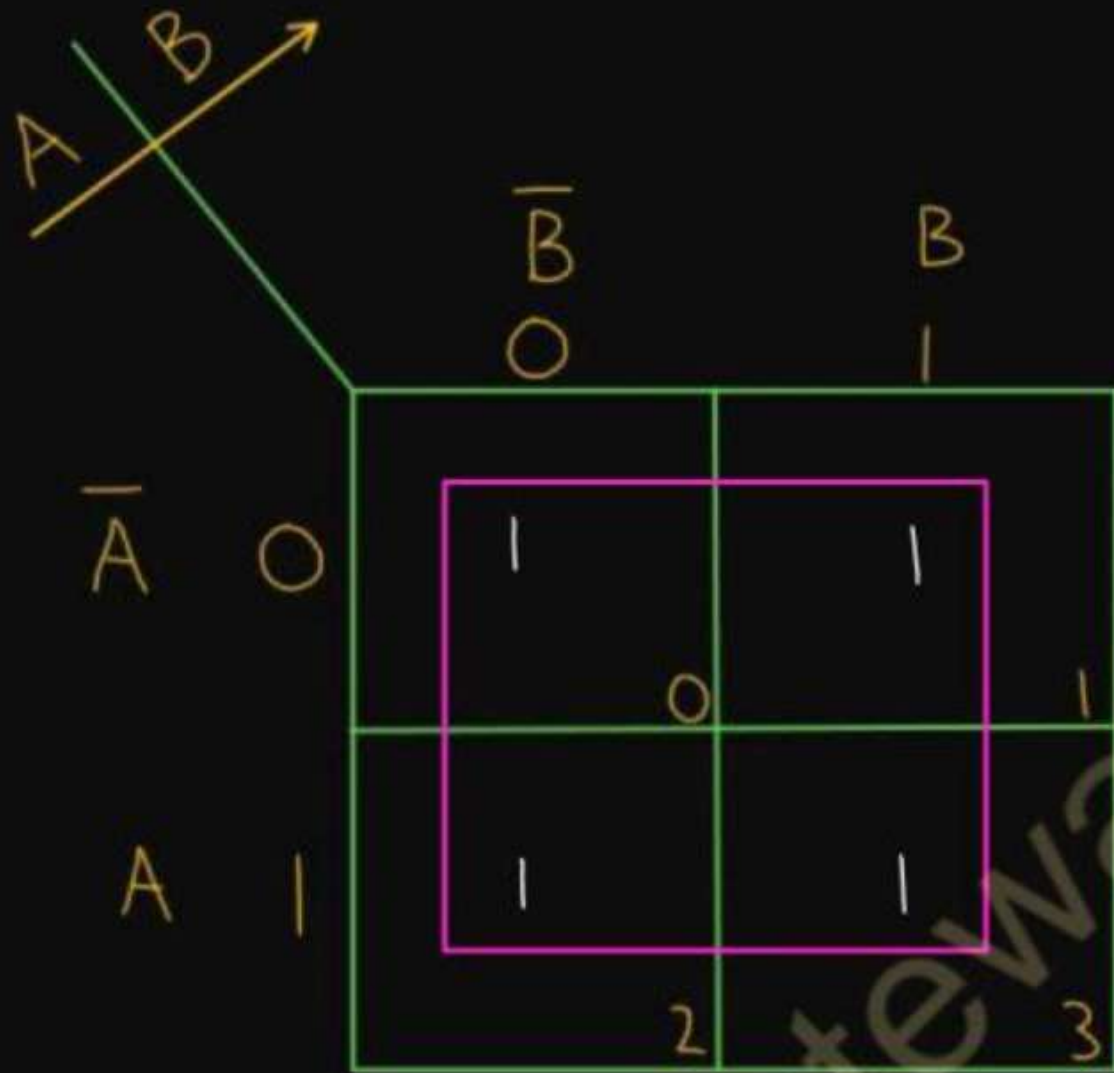
(4) 1 Group = 2^0 (0 Variables minimise)

Q.1 Simplify the following function using K-map $F(A, B) = \sum m(0, 1, 3)$



$$F(A, B) = \bar{A} + B$$

Q.2 Simplify the following function using K-map $F(A, B) = \sum m(0, 1, 2, 3)$



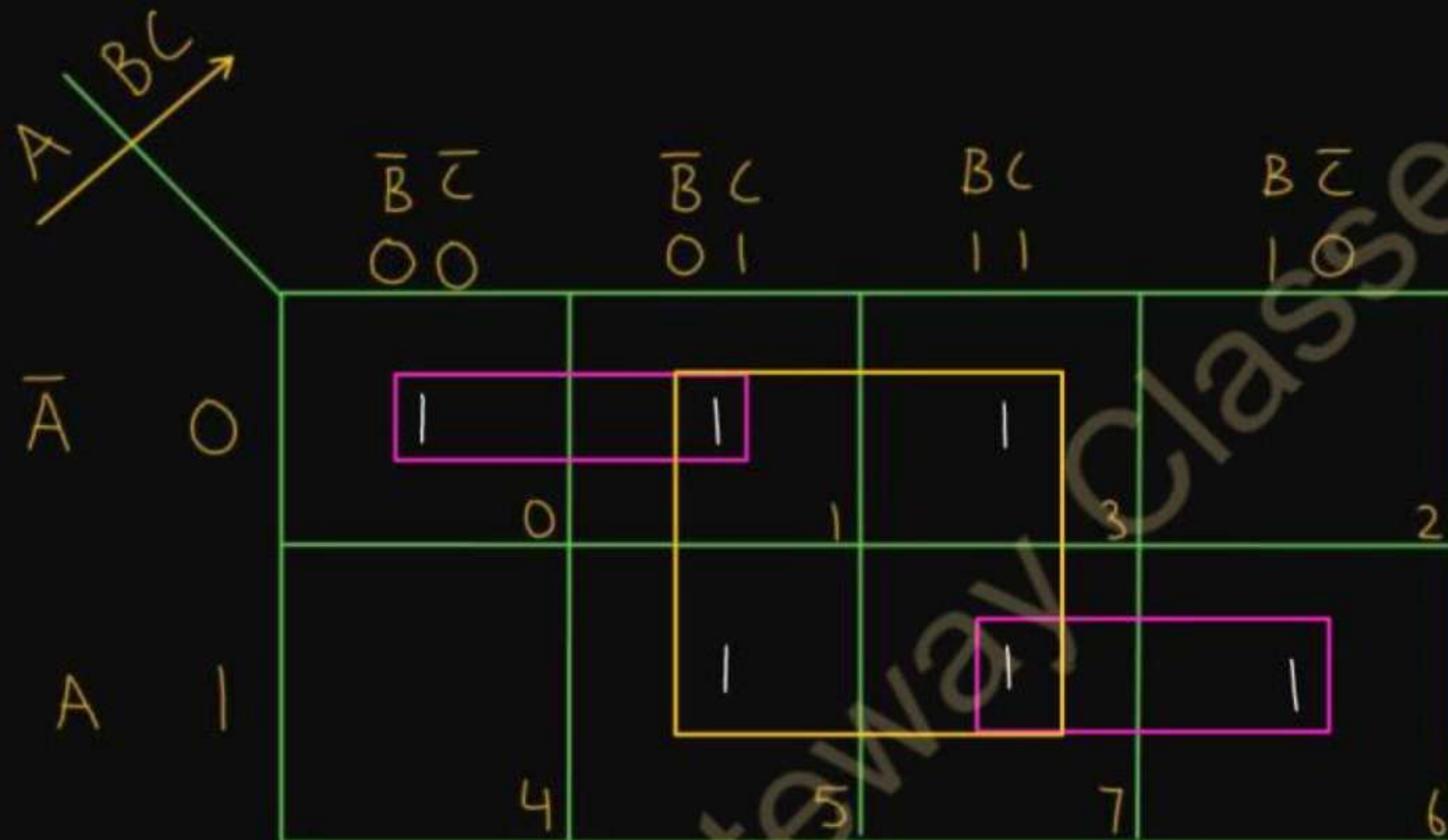
$$F(A, B) = 1$$

Q.3 Simplify the following function using K-map $F(A, B, C) = \sum m(0, 1, 3, 6, 7)$

A \ BC					
		$\bar{B}\bar{C}$ 00	$\bar{B}C$ 01	BC 11	$B\bar{C}$ 10
$\bar{A}$	0	1	1	1	
A	1			1	1

$$F(A, B, C) = \bar{A}\bar{B} + AB + BC$$

Q.4 Simplify the following function using K-map $F(A, B, C) = \sum m(0, 1, 3, 5, 6, 7)$



$$F(A, B, C) = \bar{A}\bar{B} + \bar{A}C + BC + AB$$

Q.5 Simplify the following function using K-map $F(A, B, C, D) = \Sigma(1, 3, 4, 5, 6, 7, 9, 11, 13, 15)$.

Also implement the simplified function using basic gates only.

AKTU-2024,2021

AB \ CD		$\bar{C}\bar{D}$ 00	$\bar{C}D$ 01	CD 11	$C\bar{D}$ 10
$\bar{A}\bar{B}$ 00	0		1	1	
$\bar{A}B$ 01	4	1	1	1	1
AB 11	12		1	1	
$A\bar{B}$ 10	8		1	1	
			9	11	10

$$F(A, B, C, D) = \bar{A}B + D$$

$$F(A, B, C, D) = \bar{A}B + D$$



➤ Combination of inputs on which the output may or may not depends are called Don't Care Condition

Q.6 Minimize using K-map and realize output using gates.

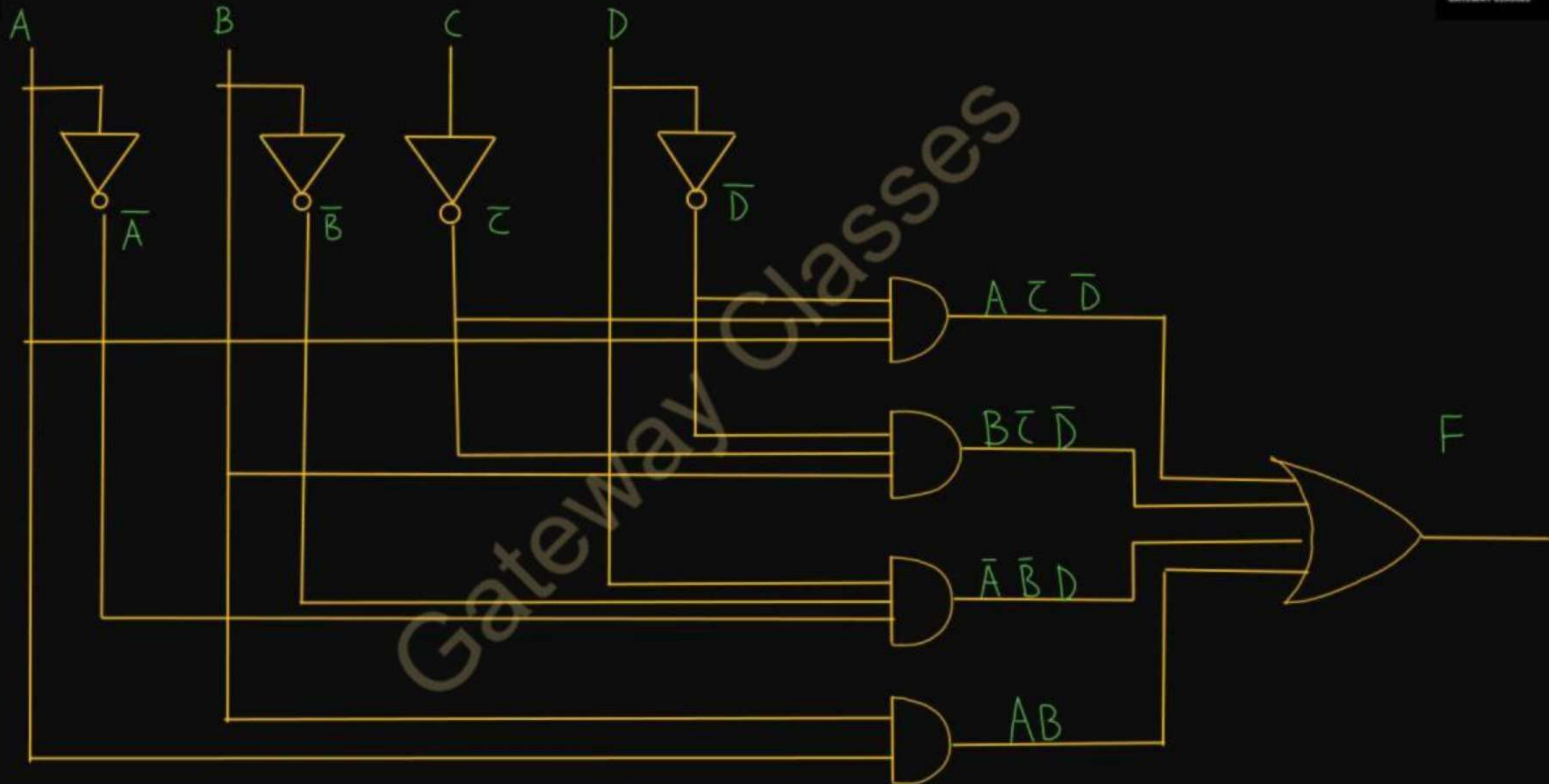
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$$F(A, B, C, D) = \sum m(1, 4, 8, 12, 13, 15) + d(3, 14)$$

AB \ CD					
		$\bar{C}\bar{D}$ 00	$\bar{C}D$ 01	CD 11	$C\bar{D}$ 10
$\bar{A}\bar{B}$	00		1	X	
$\bar{A}B$	01	1			
AB	11	1	1	1	X
$A\bar{B}$	10	1			

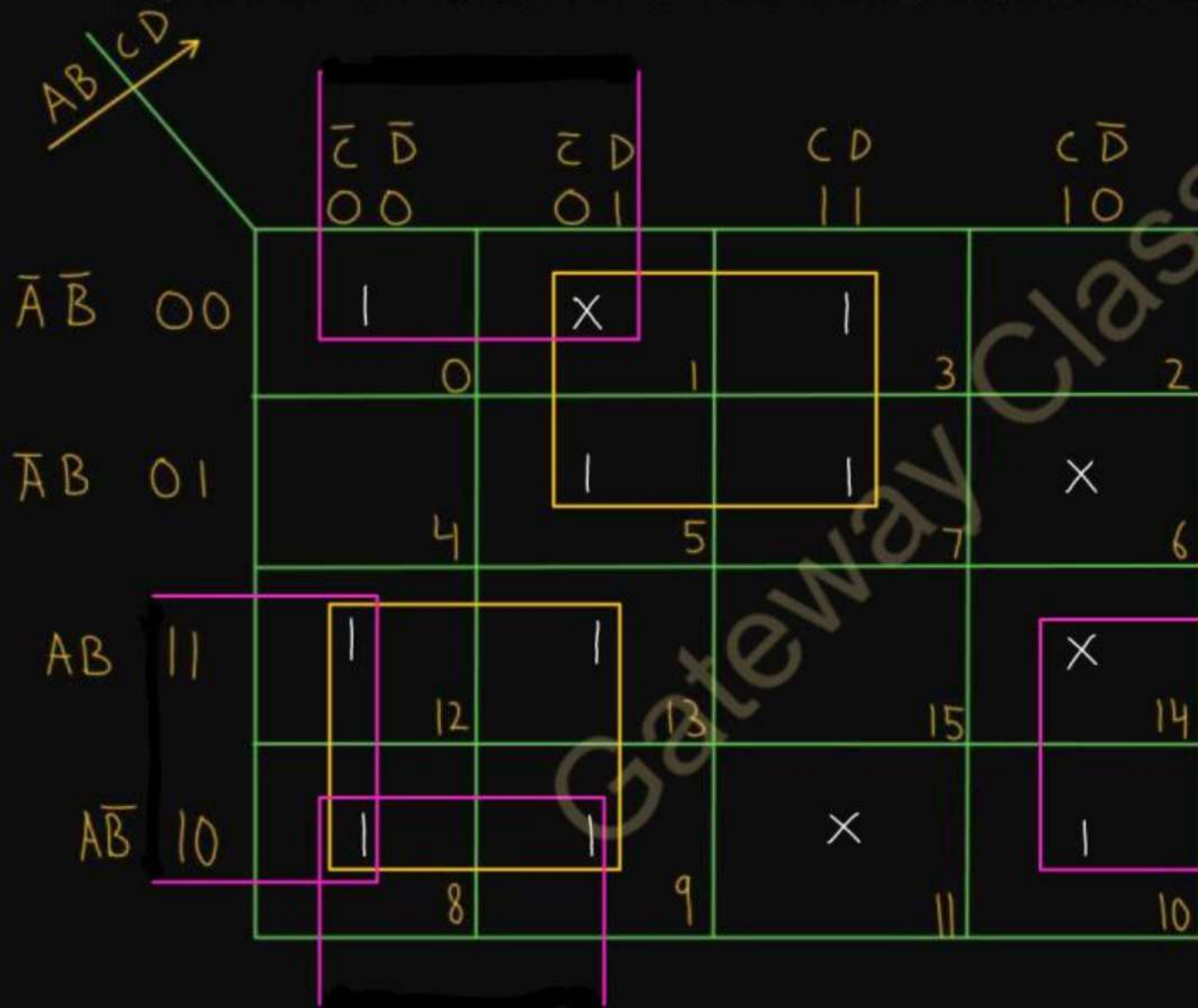
$$F = AB + \bar{A}\bar{B}D + \bar{C}\bar{D}B + \bar{C}\bar{D}A$$

$$F = AB + \bar{A}\bar{B}D + B\bar{C}\bar{D} + A\bar{C}\bar{D}$$



Q.7 Simplify the function $F = \sum m(0, 3, 5, 7, 8, 9, 10, 12, 13) + \sum d(1, 6, 11, 14)$ using K-map and implement the simplified expression using NAND gates only

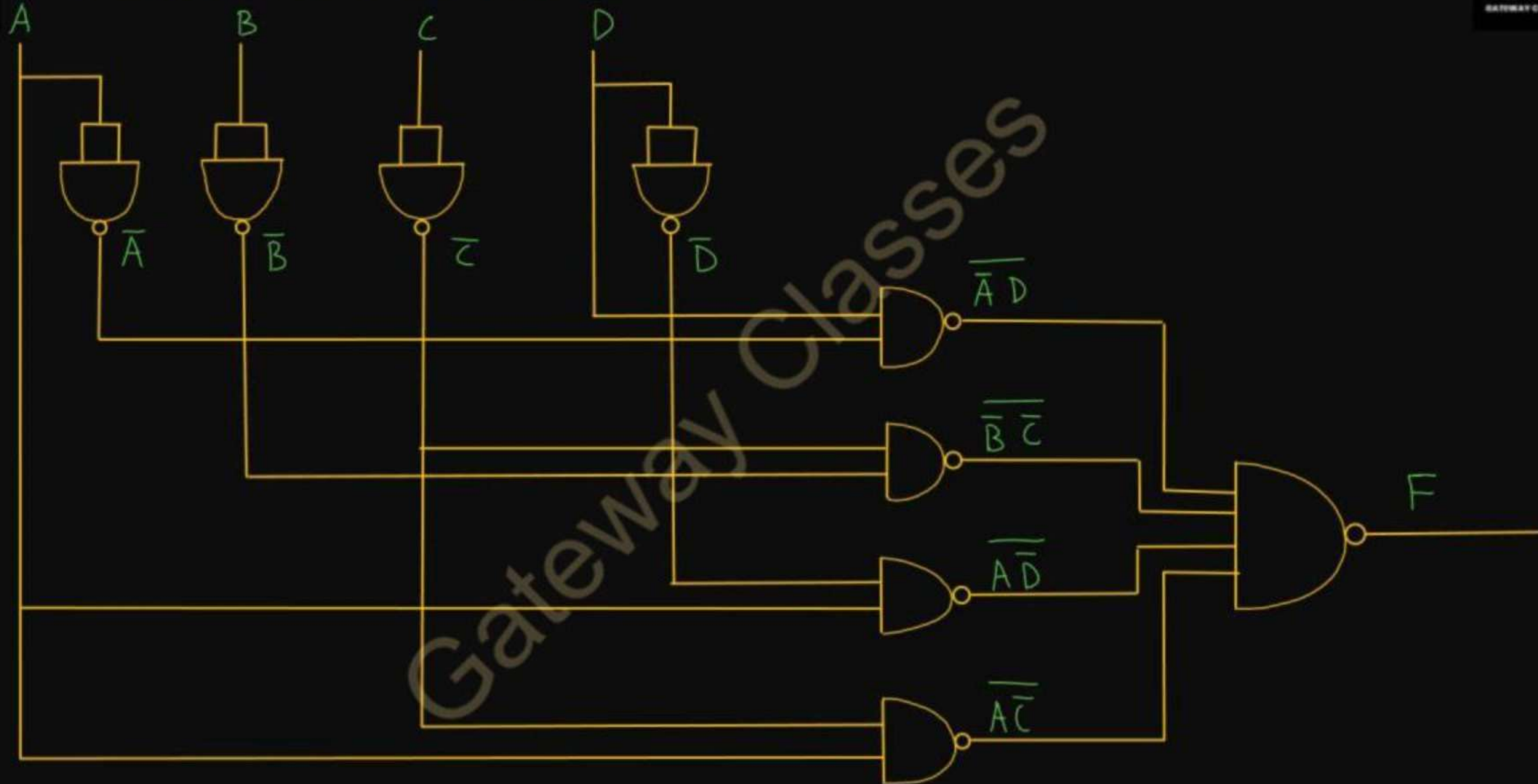
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$$F = \bar{A}D + \bar{B}\bar{C} + A\bar{D} + A\bar{C}$$

$$F = \bar{A}D + \bar{B}\bar{C} + A\bar{D} + A\bar{C}$$

$$F = \bar{A}D + \bar{B}\bar{C} + A\bar{D} + A\bar{C}$$



Q.8 Simplify the Boolean Function $F = \sum m (0,1,2,5,7,8,9,10,13,15) + d (12,14)$ using K-map and implement the simplified expression using NAND gates only. **AKTU-2023**

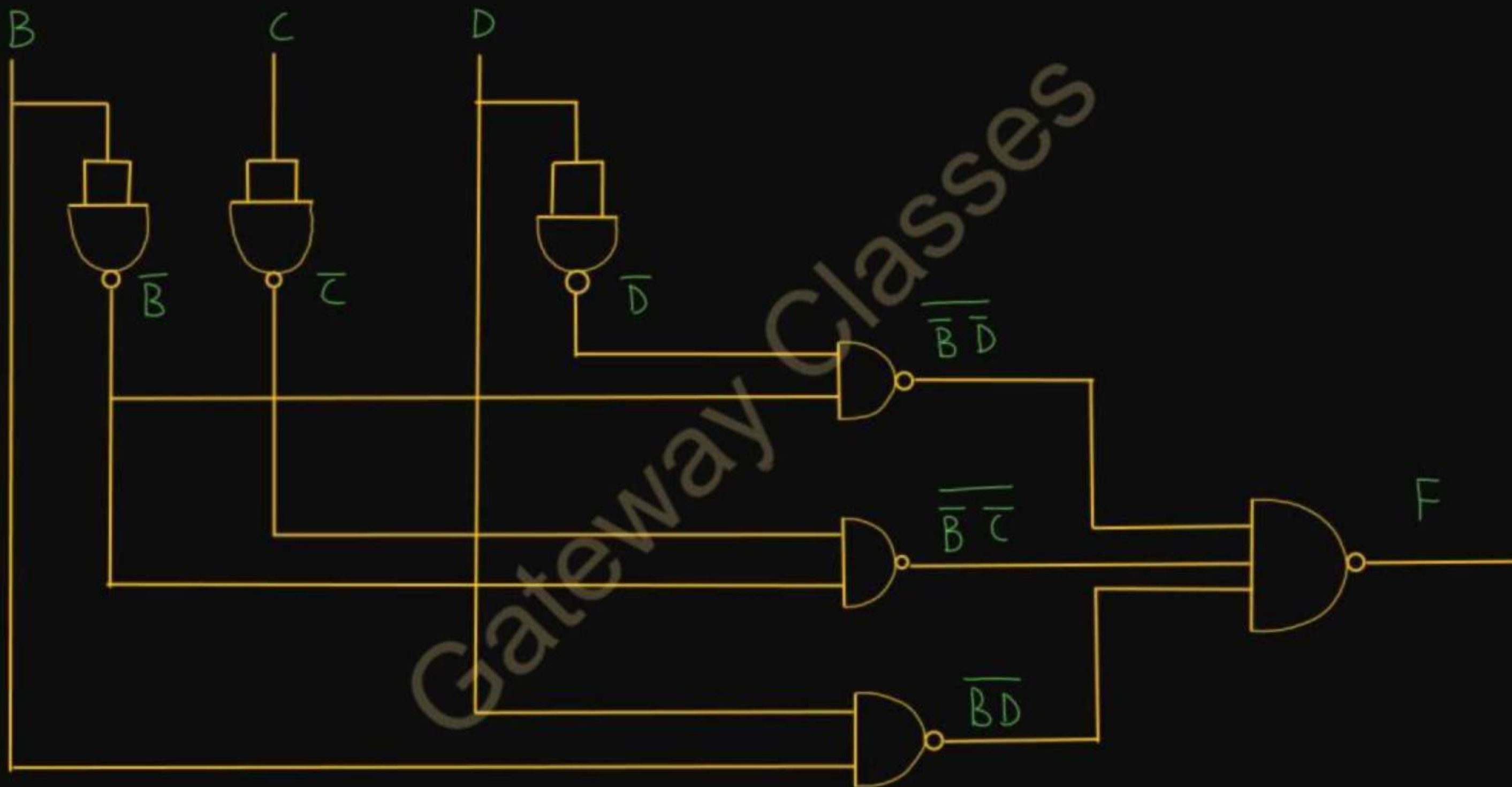
AB \ CD

$\bar{C}\bar{D}$ 00	$\bar{C}D$ 01	CD 11	$C\bar{D}$ 10
$\bar{A}\bar{B}$ 00	1	1	1
$\bar{A}B$ 01		1	
AB 11	X	1	X
$A\bar{B}$ 10	1	1	1
	0	1	3
	4	5	7
	12	13	15
	8	9	11
			10

$$F = \bar{B}\bar{D} + BD + \bar{B}\bar{C}$$

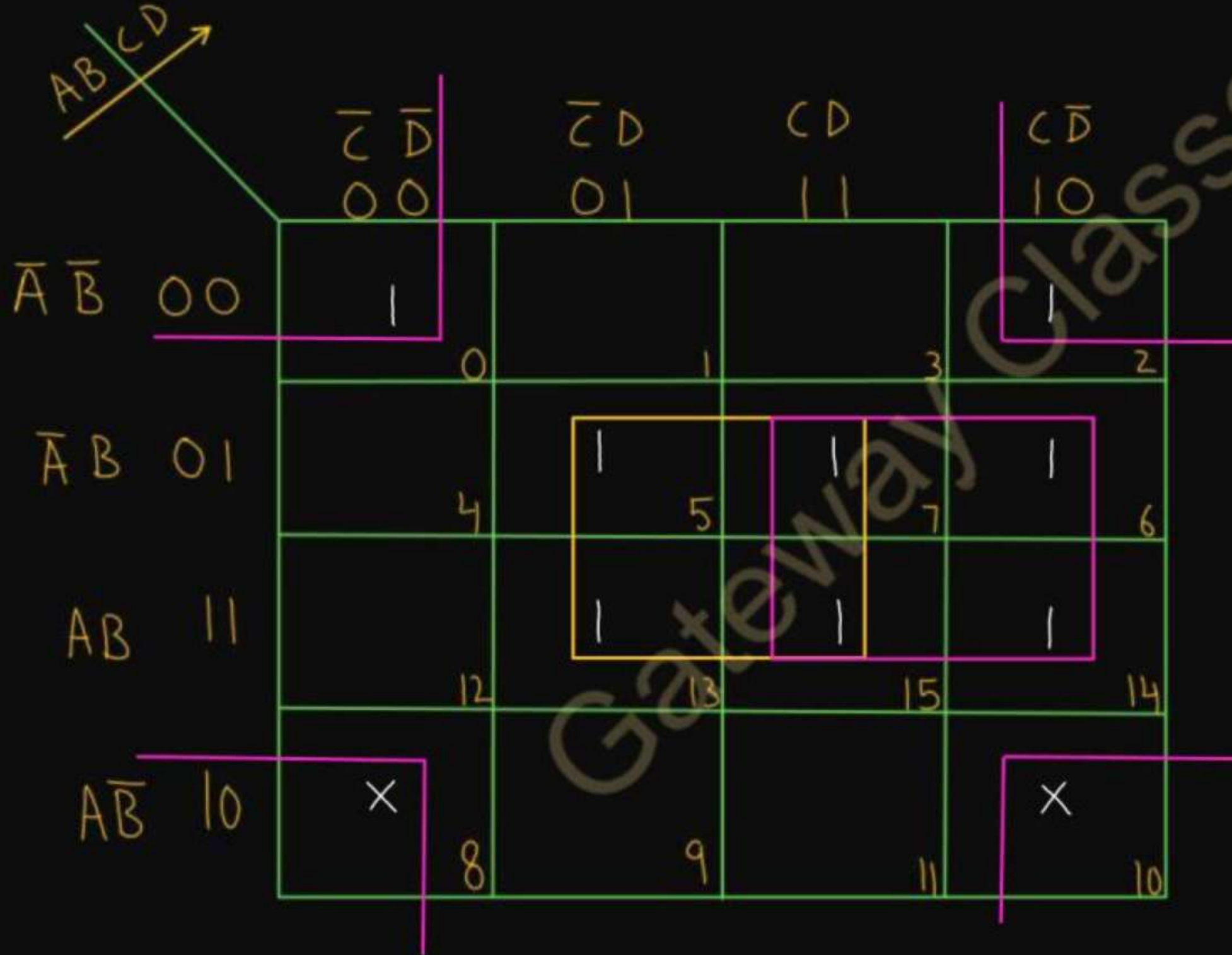
$$F = \bar{B}\bar{D} + BD + \bar{B}\bar{C}$$

$$F = \overline{B} \overline{D} + BD + \overline{B} \overline{C}$$



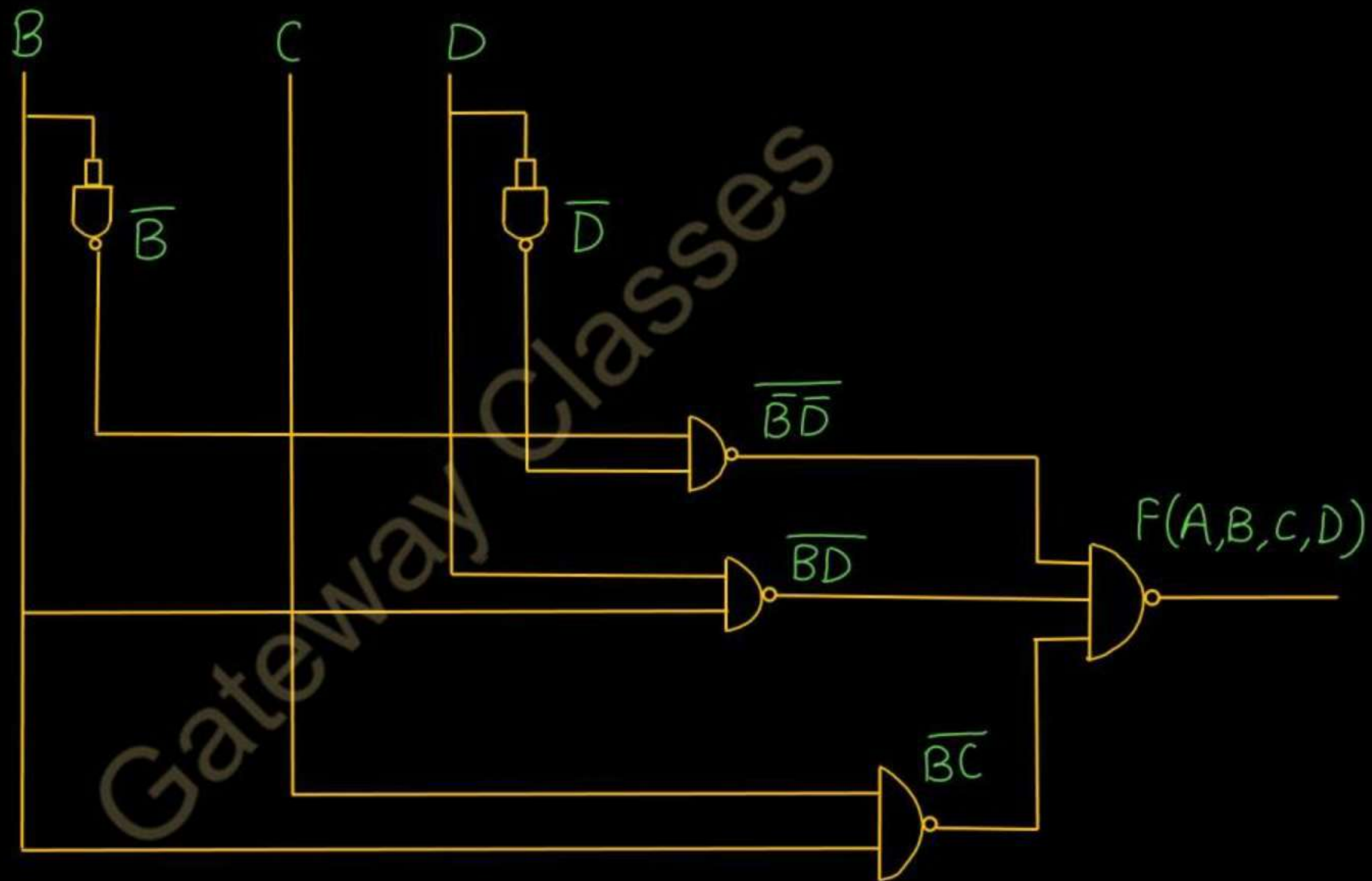
Q.9 Simplify the function $F(A, B, C, D) = \sum m(0, 2, 5, 6, 7, 13, 14, 15) + d(8, 10)$ using K-map and implement the simplified function using NAND gates only.

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$$F = \overline{B}\overline{D} + BD + BC$$

$$F = \overline{\overline{\overline{B}\overline{D} + BD + BC}}$$



Q.10 - $F(A, B, C, D, E) = \sum m(0, 1, 2, 4, 5, 6, 10, 13, 14, 18, 21, 22, 24, 26, 29, 30)$. Simplify the function with help of

K-map and realize the simplified function using basic logic gates.

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$\overline{A}$

BC \ DE

	$\overline{D}\overline{E}$ 00	$\overline{D}E$ 01	DE 11	$D\overline{E}$ 10
$\overline{B}\overline{C}$ 00	1	1		1
$\overline{B}C$ 01	1	1		1
BC 11		1		1
$B\overline{C}$ 10				1

A

BC \ DE

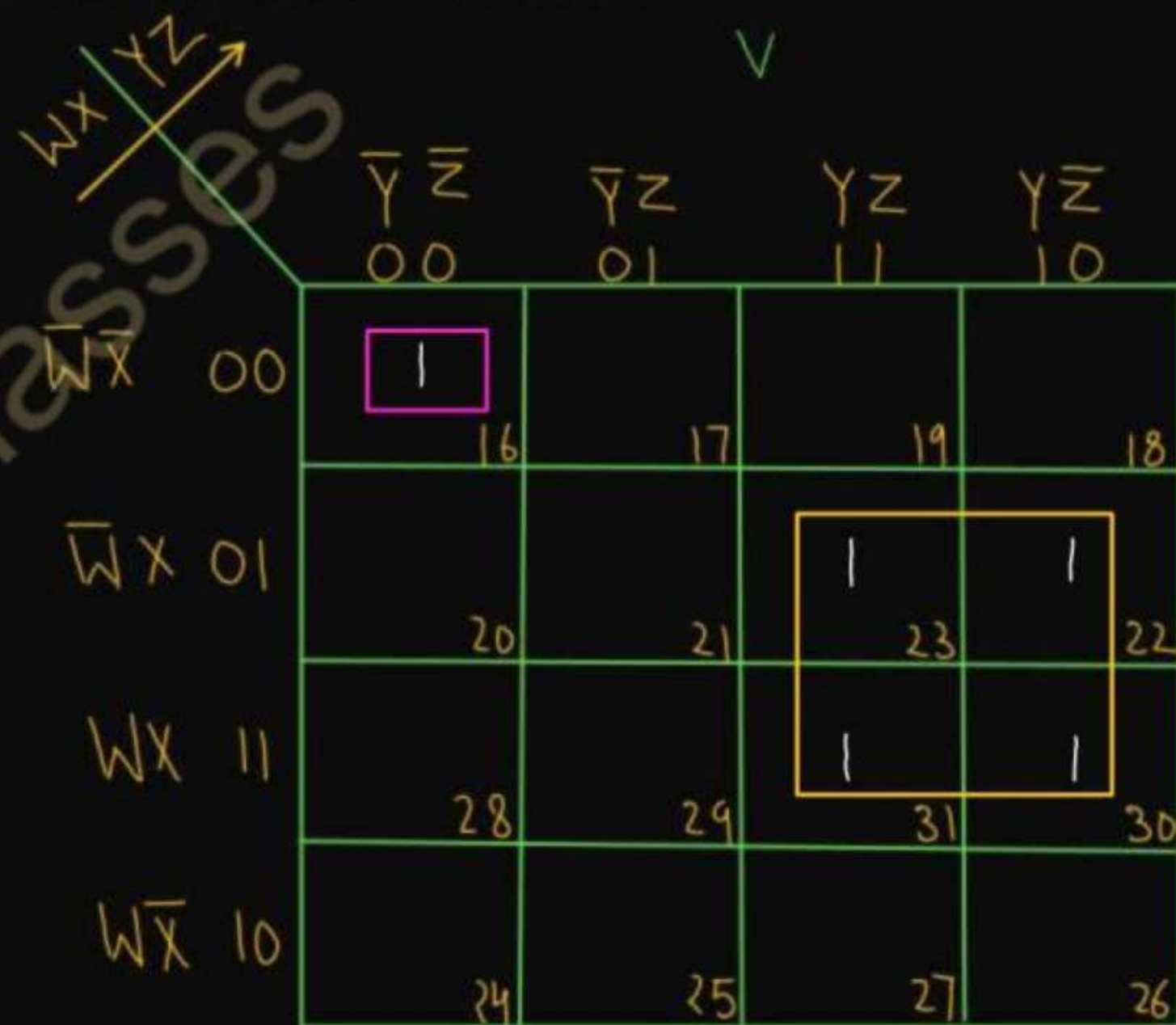
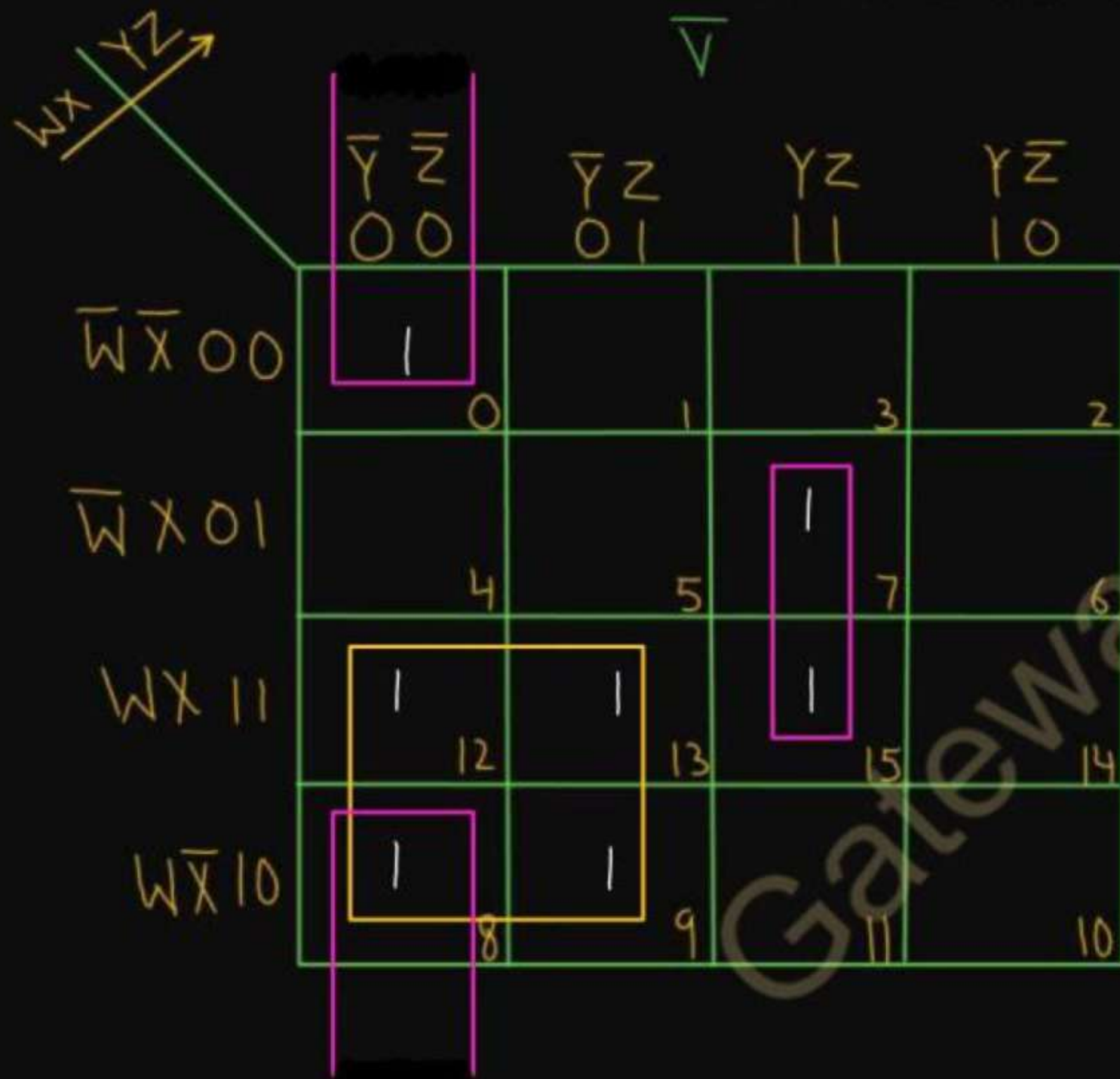
	$\overline{D}\overline{E}$ 00	$\overline{D}E$ 01	DE 11	$D\overline{E}$ 10
$\overline{B}\overline{C}$ 00				1
$\overline{B}C$ 01		1		1
BC 11		1		1
$B\overline{C}$ 10	1			1

$$F = D\overline{E} + \overline{A}\overline{B}\overline{D} + C\overline{D}E + AB\overline{C}\overline{E}$$

Q.11 Simplify the following Boolean expression using K-map.

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$$f(V, W, X, Y, Z) = \sum m(0, 7, 8, 9, 12, 13, 15, 16, 22, 23, 30, 31)$$



Q.12 Simplify the following Boolean expression using K-map.

Ans

$$F = \bar{D}\bar{F} + \bar{A}\bar{B}C\bar{E} + \bar{A}C\bar{E}F + BCD\bar{F} + A\bar{B}EF$$

$f(A, B, C, D, E, F) = \sum m(0, 2, 8, 9, 10, 12, 13, 16, 18, 24, 25, 26, 29, 31, 32, 34, 35, 39, 40, 42, 43, 47, 48, 50, 56, 58, 61, 63)$

$\bar{A}$

		$\bar{B}$			
	CD \ EF	00	01	11	10
00		1 0	1	3	2
01		4	5	7	6
11		1 12	1 13	15	14
10		1 8	1 9	11	10

B

		B			
	CD \ EF	00	01	11	10
00		1 16	17	1 18	19
01		20	21	23	22
11		28	1 29	1 31	30
10		1 24	1 25	27	1 26

A

		A			
	CD \ EF	00	01	11	10
00		1 32	33	1 35	1 34
01		36	37	1 39	38
11		44	45	1 47	46
10		1 40	41	1 43	1 42

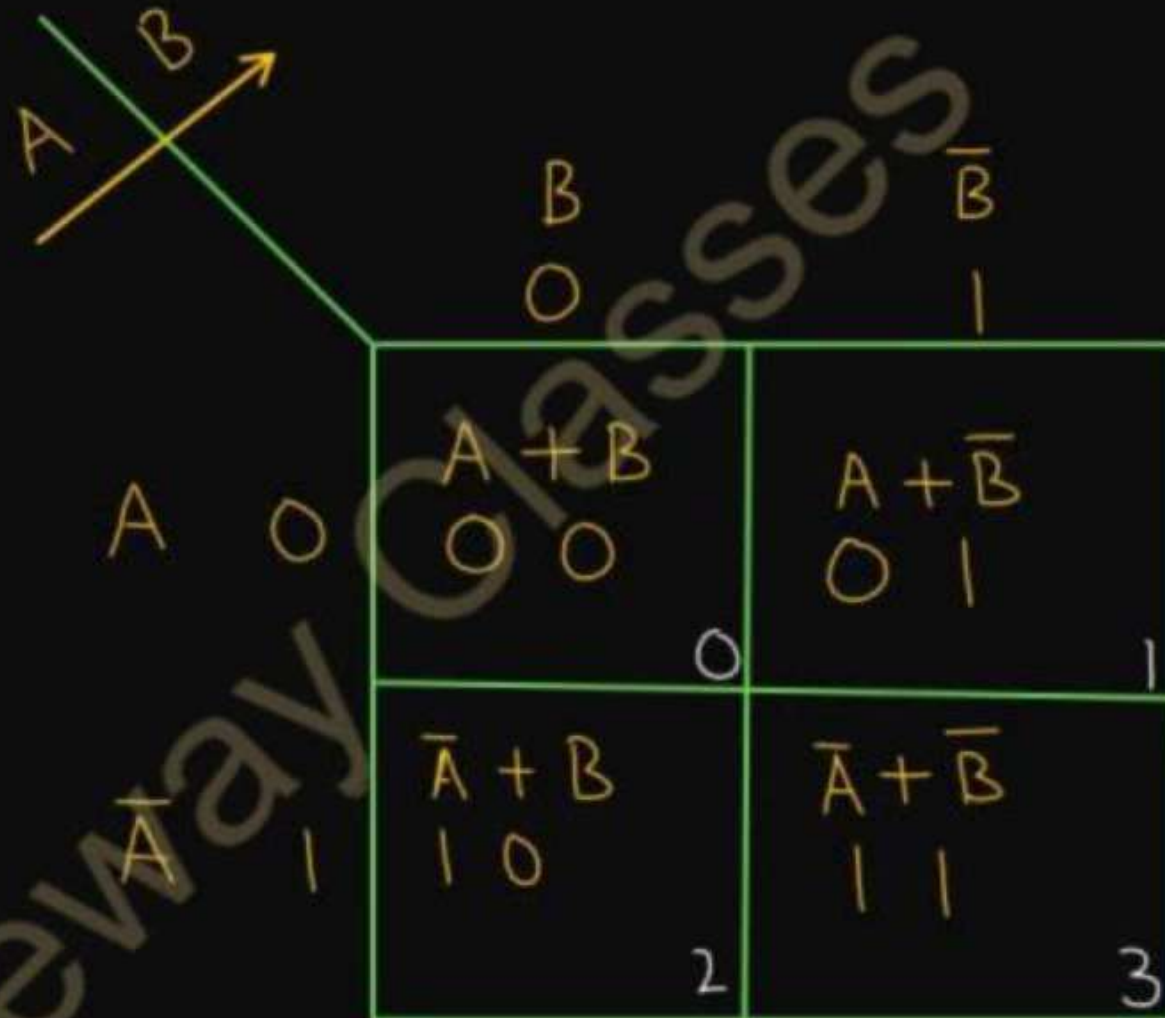
$\bar{A}$

		$\bar{A}$			
	CD \ EF	00	01	11	10
00		1 48	49	51	1 50
01		52	53	55	54
10		60	1 61	1 63	62
11		1 56	57	59	1 58

K-Map for POS Form

(1) For two variable A and B

- No of cells in K-map = $2^2 = 4$
- Each cell contain a Maxterm



(2) For three variables A, B, C

- No of cells in K-map = $2^3 = 8$
- Each cell contain a Maxterm

		BC			
		$B+C$ 00	$B+\bar{C}$ 01	$\bar{B}+\bar{C}$ 11	$\bar{B}+C$ 10
A	0	$A+B+C$ 000 0	$A+B+\bar{C}$ 001 1	$A+\bar{B}+\bar{C}$ 011 3	$A+\bar{B}+C$ 010 2
	1	$\bar{A}+B+C$ 100 4	$\bar{A}+B+\bar{C}$ 101 5	$\bar{A}+\bar{B}+\bar{C}$ 111 7	$\bar{A}+\bar{B}+C$ 110 6

(3) For three variables A, B, C, D

- No of cells in K-map = $2^4 = 16$
- Each cell contain a Maxterm

ABCD

	$C+D$ 00	$C+\bar{D}$ 01	$\bar{C}+\bar{D}$ 11	$\bar{C}D$ 10
$A+B$ 00	0	1	3	2
$A+\bar{B}$ 01	4	5	7	6
$\bar{A}+\bar{B}$ 11	12	13	15	14
$\bar{A}+B$ 10	8	9	11	10

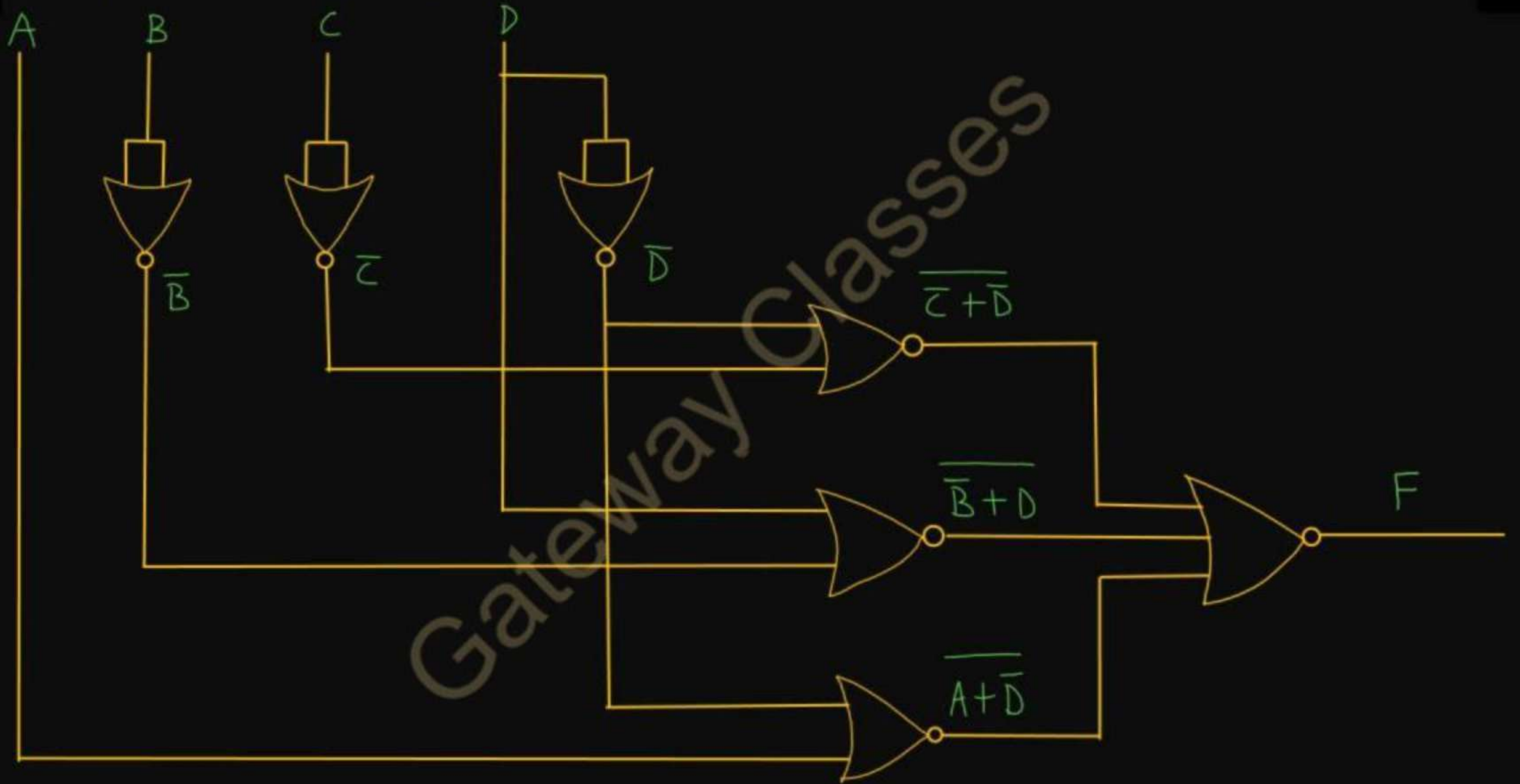
Q.1 Simplify the Boolean expression using K-map and implement the simplified expression using NOR GATES only $F(A, B, C, D) = \prod M(1, 3, 4, 5, 6, 7, 11, 12, 14, 15)$ **AKTU-2023**

AB \ CD					
		00	01	11	10
A+B	00		0	0	
	01	0	0	0	0
$\bar{A} + \bar{B}$	11	0		0	0
	10			0	

$$F = (\bar{B} + D)(A + \bar{D})(\bar{C} + \bar{D})$$

$$F = \overline{(\bar{B} + D)(A + \bar{D})(\bar{C} + \bar{D})}$$

$$F = (\overline{B} + D)(A + \overline{D})(\overline{C} + \overline{D})$$



Q.2 Simplify the function $F = \prod M(0, 2, 10, 11, 12, 14, 15) \cdot \prod d(5, 13)$ using K-map and implement the simplified expression using NOR gates only.

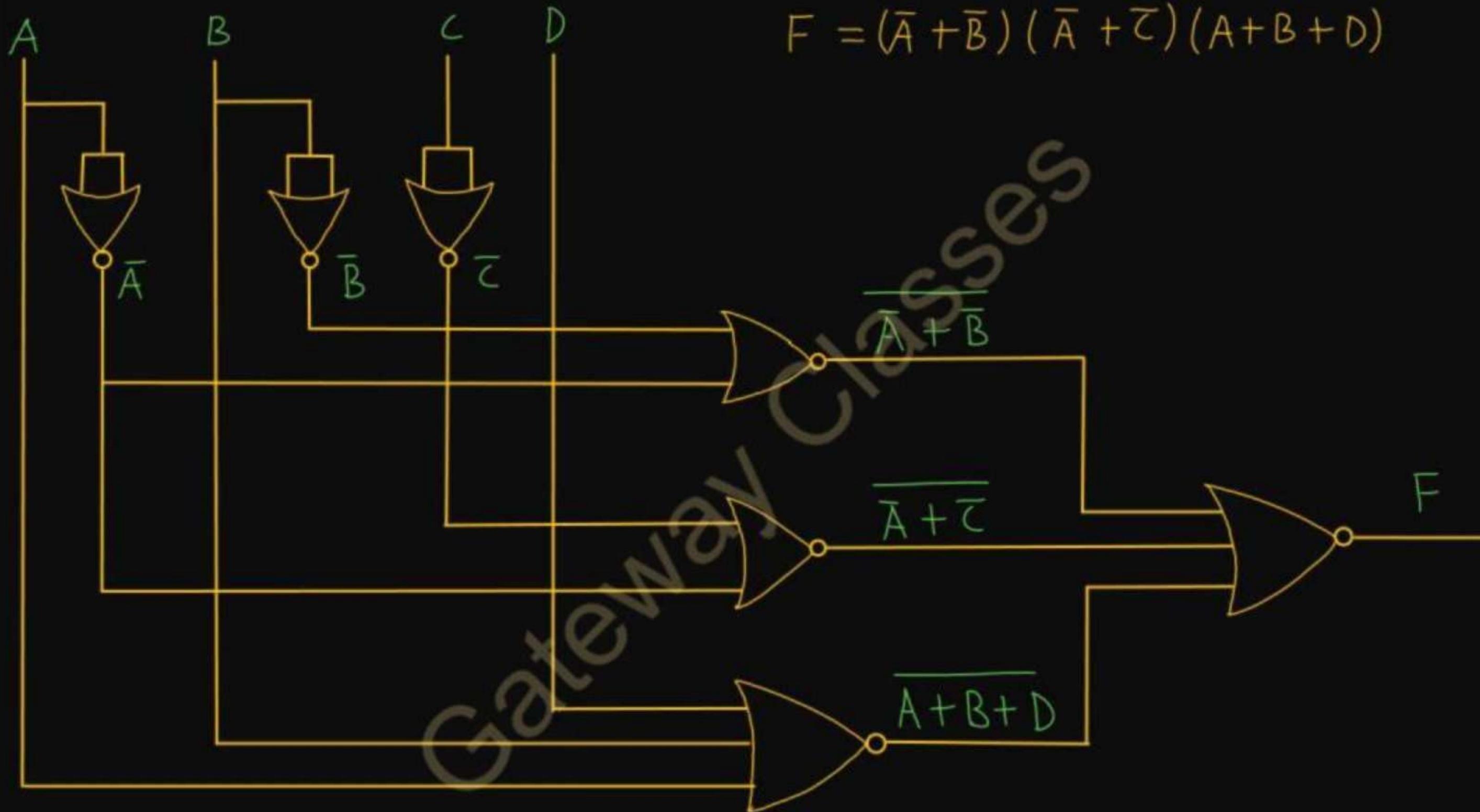
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AB \ CD					
		CD	C $\bar{D}$	$\bar{C}\bar{D}$	$\bar{C}D$
A+B	00	0			0
	01		X		
$\bar{A}+\bar{B}$	11	0	X	0	0
	10			0	0

$$F = (\bar{A} + \bar{B})(\bar{A} + \bar{C})(A + B + D)$$

$$F = \overline{(\bar{A} + \bar{B})(\bar{A} + \bar{C})(A + B + D)}$$

$$F = (\bar{A} + \bar{B})(\bar{A} + \bar{C})(A + B + D)$$



$$F(A, B, C, D) = \Pi M(3, 4, 5, 7, 9, 13, 14, 15). d(0, 2, 8).$$

AB \ CD					
		00	01	11	10
A+B	00	x		0	x
	01	0	0	0	
$\bar{A} + \bar{B}$	11		0	0	0
	10		0		

$$F = (A + \bar{B} + C)(A + \bar{C} + \bar{D})$$

$$(\bar{A} + C + \bar{D})(\bar{A} + \bar{B} + \bar{C})$$

UNIT -4 : DPP

Topic : Digital Electronics

Q.1 Define the term "combinational logic" in digital electronics.

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Q.2 Add binary numbers $(1110.10 + 1011.11)_2$.

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Q.3 Calculate the 1's and 2's complement of binary number 1010101.

AKTU-2023

Q.4 Determine base of the following: (i) $(345)_{10} = (531)_x$ (ii) $(2374)_{16} = (9076)_x$

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Q.5 Write the truth table of two input X-OR gate and two input X-NOR gate.

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Q.6 Find 1's and 2's complement of : 1101001

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Q.7 Perform the following as mentioned:

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(i) Convert $(63.250)_{10}$ to binary number.

(ii) Convert $(10010.101)_2$ to decimal number.

(iii) Convert $(A6B.0F)_{16}$ to octal number.

(iv) Perform subtraction using 2's complement $(111-1010)_2$.

(v) Design AND, OR, and NOT gates using only NOR gates.

Q.8 Convert the following:

- (i) $(53.625)_{10}$ to $(?)_2$
- (ii) Find the base x if $(211)_x = (152)_8$
- (iii) Subtract using 1's complement: $(10111)_2 - (110011)_2$
- (iv) find the 1's and 2's compliment of $(010100)_2$

Q.9 Perform following operation as indicated.

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- (i) Determine 2's complement of $(1010.110)_2$.
- (ii) Convert $(25.125)_{10}$ into Hexadecimal number.
- (iii) Add binary number $(1011)_2$ and $(1111)_2$.
- (iv) State De Morgan's Law.
- (v) Define minterm and maxterm.

Q.10 (i) Subtract using 10's complement: $(9754)_{10} - (364)_{10}$

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(ii) Subtract using 1's complement: $(10111)_2 - (110011)_2$



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